

Of Bytes and Watts: Regulatory Safeguards of Powering AI Infrastructure

Using network modelling of international energy trade and causal
analysis of data centre regulation frameworks to inform policy insights
that aim to manage data centre growth

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Executive Summary

Data centres are an increasingly critical part of the global economy. They play an essential role in powering Artificial Intelligence (AI) models, storing vast amounts of data, and providing solutions related to cloud computing. At the same time, they place an intensive demand on resources such as energy, water, and land. To match this demand, utilities often expand or modify the electricity grid, which can be an expensive process. The additional costs incurred in establishing such infrastructure are often offset by households and other customers who may not directly reap the benefits of these upgrades. Furthermore, data centre presence has been found to drive up local electricity prices for all consumers. This leads us to the question of *fairness* in the cost allocation of energy and infrastructure to support data centres, and whether it is fair for households to be bearing the brunt of the steep costs associated with powering data centre infrastructure. To address this, many countries have developed regulatory frameworks meant to mitigate the potentially harmful effects of such infrastructure.

The objective of this report is to study how the growing presence of data centres relates to a range of variables, including electricity price, international energy trade, energy generation mix (fossil, renewable, nuclear), geographical factors (population, land area), and different categories of regulatory frameworks (planning, reporting, sustainability, and incentives). The research problem is multi-faceted, involving several factors that are possibly intertwined. The complexity of these relationships makes it important to distinguish correlation and causality, as seemingly meaningful relationships could be spurious or directionally complicated. Furthermore, countries do not exist in a vacuum; their relationships with each other (such as how they trade electricity) are important to consider.

In order to uncover these relationships, we adopted a four-part methodology, which all come together to provide a holistic view of our research problem. The first part is based on a careful study of regulatory frameworks implemented by different countries and organisations (such as the European Union) that were directed at data centres or similar infrastructure. Based on this, four broad categories of regulations were identified: energy reporting, sustainability, zoning restrictions (i.e. planning), and incentives. The regulations, policies, directives, and binding frameworks related to these categories were sourced and studied for 45 European and Eurasian countries. A criterion was developed to ordinaly score each of these countries on the different policy categories, on a scale of 0 to 3. They were then accordingly hand-coded using a label encoding process. This formed a key component of the data for the subsequent models. We endeavoured to keep the criteria objective and measurable, but since some decisions required discretion, we note a construct-validity limitation.

The second part of our methodology section comprises the exploratory analysis of data from all 51 US states. We used regression analysis to test the relationship between data

centre presence and state-level energy prices. As our two key variables of interest, we felt it was important to begin with this step, testing their effect on one another. The US context provides a greater sample size ($n = 51$ as opposed to our 29 European countries studied later), so serves as a more statistically reliable context to analyse the relationship between price and data centres. We do, however, acknowledge that results throughout this report, given our two separate geographical contexts and the apparent importance of physical and jurisdiction-specific factors in our research question, may be location-specific, and so are interpreted with this in mind.

The third part of our methodology involves the implementation of graph modelling and network analysis. We built graph objects in *R* which represent electricity trade (the *edges*) among 29 European countries from 2023 to 2025. Each country (*node*) is attached to a set of common attributes including data centre count and energy prices, alongside our other relevant variables, and analysed these in a pipeline of three separate techniques.

Firstly, Exponential Random Graph Modeling (ERGM) was employed to assess the relationship of our key variables with the presence of a trade tie between countries. ERGMs estimate which structural and nodal features are associated with tie presence (the outcome variable) in the observed network, but since our graphs are single-year, static snapshots, these associations cannot establish a causal direction. In our case, we believe that *reverse-causality* (network structure influencing data centre presence and price, rather than the reverse) is *more* likely, or at least equally plausible, since trade ties are established and embedded in fixed infrastructure, whereas data centre presence and energy prices are comparatively variable over the same period.

This is followed by cross-sectional regressions of each year’s network separately. These specifications included engineered network exposure variables designed to encode the structural features of the networks that emerged significant from the ERGM. This method operationalises the results of the ERGM by constructing each country’s average price gap with its trading partners and testing its association with data centre presence via OLS regression.

The final stage of the network analysis pipeline is a Network Autocorrelation Model, which tests whether countries’ variables move in correlation once network dependence is modelled formally, rather than approximated by an exposure variable. The results of these methods come together to provide an overview of the effects observed when our observations (countries) are modelled as relationally dependent (each country’s outcomes linked to its trading partners’, not just serially correlated over time) on one another. Results are interpreted cautiously, given relatively small ($n = 29$) sample size.

The fourth methodological approach in this report revolves around causal analysis. A modern inference framework called DoWhy charted out a pipeline for causal inference, consisting of four steps: Model, Identify, Estimate, and Refute. Using the Linear Non-Gaussian

Acyclic Model (LiNGAM) for causal discovery, a Directed Acyclic Graph (DAG) was built, which provided the basis for causal understanding. Following this, the Average Treatment Effect (ATE) of features such as regulatory strength were tested on data centre presence using the Backdoor Estimation Criterion, with additional tests to ensure the estimates were trustworthy. These causal effects were used to inform a cross-sectional regression model, built using the causally significant features.

The main practical implication of our analyses as a whole is that policy design appears to matter for data centre growth, particularly planning frameworks and incentive schemes, which emerge as the clearest causal levers. Network exposure regressions found incentives regulation to be a consistently significant factor with a positive effect on data centre presence ($p = 0.003 - 0.018$). The most extreme effect is driven solely by Ireland; however, but our small sample size means this does not invalidate the finding, but rather warrants further research. Cross-sectional causal-aware regressions all confirmed the positive significant effect of incentives regulation. Planning regulation appears significant ($p = 0.037$) in its DoWhy causal estimand, similarly exhibiting a positive effect on the presence of data centres. Causal analysis identifies sustainability regulation as relevant, though this is more indirect, intertwined with broader institutional context. These findings are directly relevant to policymakers seeking to attract data centre investment or manage its spatial and infrastructural consequences.

The ERGM shows a link between data centre presence and energy trade ties and a highly significant relationship between trade and price differences between countries. The Network Autocorrelation model confirms that countries' prices are not independent of one another ($LR = 12.9 - 21.6$, $p < 0.001$), whilst US data linear regressions showed that price is a significant predictor of data centres ($p = 0.011 - 0.012$). Although the coefficient sign flips upon controlling for population via a per-capita data centre measure, the significance of this relationship is the key insight here. Holistically, these findings indicate that countries linked by energy trade affect each other's prices, which we found separately to have a relationship with data centres; indirectly linking our key variables. The causal mechanisms involved in this warrant further empirical study, presenting opportunity for further research.

This is not the only avenue for further research that our study opens up. Future work could build on our causal analysis to employ double machine learning or causal forests to capture non-linear and heterogeneous treatment effects while retaining DAG-informed adjustment sets. On the data side, extending the panel to a longer time horizon and incorporating more granular measures of variables would enable richer causal identification, possibly including the use of instrumental variables based on exogenous shocks or historical policy discontinuities. Crucially, further study considering a wider global scope would aid in a better holistic understanding of this highly complex and fast-growing problem.

The code and analysis for this study has been made available here

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1 Introduction

In recent years, the expansion of hyper-scale data centres has placed unprecedented pressure on electricity systems (*International Energy Agency* 2025). Gigawatt-scale clusters have emerged, with thriving Artificial Intelligence increasingly demanding larger and more power-intensive data centres (*International Energy Agency* 2025). In some jurisdictions, this has been linked to rising electricity prices, leading to low public acceptance. Understanding how the data centre pipeline is evolving is therefore critical for policymakers and regulators. But data centres are often portrayed as essential drivers of economic growth, including job creation and localised economic development (Alvarez et al. 2026), which drives mounting pressure on policymakers to design policies that facilitate data centre investments. On the other hand, the necessary grid investments can be seen in some cases as benefitting a limited subset of grid users, while their costs are distributed across the entire consumer base (Pescoe & Martin 2025).

This is precisely the problem that forms the basis of our research as we aim to uncover a set of data-driven, robust insights that might help to *inform* policymakers on how best to tackle the complex issue of regulating data centres amidst the rising urgency and compounding pressure. Given the highly complex nature of the scope of this research, we do not expect to formulate policy *recommendations* from this analysis alone. However, we hope to identify areas of focus for further study that would be key in informing policymakers when it comes to regulating data centres to ensure fairer cost allocation of energy.

Ultimately, our findings indicate that policy design matters for data centre growth, with sustainability, planning and incentive frameworks emerging as the clearest causal levers. Incentives regulation was consistently associated with a positive effect on data centre presence across the network exposure regressions and all cross-sectional causal-aware regressions, while planning and sustainability regulations also showed significant positive effects in its DoWhy causal estimand. Alongside this, we found that European countries' energy prices cannot be considered in isolation from their trading partners. Exponential Random Graph Modelling revealed a significant link between data centre presence and energy trade ties, and a highly significant relationship between trade and cross-country price differences, while the Network Autocorrelation model confirmed that countries' prices are not independent of their trading partners'. In the US context, price was itself a significant predictor of data centre presence, though this relationship's direction depends on whether population is controlled for.

Our approach takes various complementary angles grounded in modern data science techniques; namely, regulation factor encoding, network modelling, and causal analysis, with the Methodology section of this report following this outline. This is preceded by an in-depth review of the relevant literature and current data centre regulation, and we end the report with a conclusion summarising our key findings.

2 Literature and Regulation Review

Data centres are computing service infrastructure that house various systems for computation and storage. This equipment is essential to “collect, store, process, and distribute” massive volumes of data for a wide range of applications as per Zhang et al. (2021). The data centre industry is a vital part of Europe’s economy, with estimates showing that colocation data centres could contribute 138 million towards GDP annually, while potentially creating over 300,000 jobs by 2031 according to the European Data Centre Association (2026).

The current state of data centres is also linked to trends in Artificial Intelligence (AI). Over the last few years, the generative Artificial Intelligence market has seen rapid growth, increasing by about 40% year by year, and is estimated to reach \$1 trillion by 2032 according to Lee & West (2025). With the rise of generative AI models and the growing demand for various AI applications, there is also a simultaneous demand for hardware and compute that can support their development, as pointed out by de Vries-Gao (2026). Data Centres are necessary for developing AI applications across various stages, including training, inference, and storage.

According to a report by the Clark et al. (2026), data centres currently consume approximately 2.5% of electricity in the United Kingdom, but this electricity consumption is expected to rise fourfold by 2030. Schneider et al. (2024) estimated that Europe’s data centre power consumption could increase by 10 – 15% over the course of the next 10 years.

Particularly relevant here is the training of Large Language Models, which have a massive number of parameters and therefore consume a high amount of electricity. Parameters here refer to the various weights and biases in an LLM that map its input to the output. Generally, as models grow more complex, they often have a higher number of parameters, which in turn leads to an increase in electricity consumption.

Jegham (2025) highlighted various energy consumption benchmarks for different models. For example, the training of GPT-3, released in 2020, took up approximately 1287 MWh. More recently, global electricity demand from data centres including those used to power AI grew by 17% in 2025. This growth is projected to continue, with electricity consumption from data centres roughly doubling from 485 TWh in 2025 to 950 TWh by 2030, reaching around 3% of global electricity demand (International Energy Agency 2026).

According to the European Data Centre Association (2026), there is set to be a projected 17% Compound Annual Growth in power demand for IT purposes in Europe, with 67% of data centre operators pointing to power access as the biggest challenge facing them. Thus, based on these various estimates, it is clear that the demand for energy due to data centres is likely to increase significantly in the coming years.

This in turn leads to another problem. In order to match this demand from data centres, electricity companies (also referred to as utilities) are required to make various additions and extensions to the existing grid. As per Nguyen & Green (2025), data centres can cause a spike in local electricity rates by “driving up overall energy demand” that would in turn put

the grid capacity under strain and “force utilities to invest in costly infrastructure upgrades”.

In many instances, utilities pass on additional infrastructure costs to other consumers through steeper monthly bills and rate increases. A Bloomberg News (2025) report found that monthly electricity costs had risen by up to 267% over five years in some US cities near clusters of data centres.

Alvarez et al. (2026) find that while the presence of data centres has some positive effects on the local economy (such as higher employment), they also drive up local electricity prices and house prices. A study by Wang et al. (2025) employed sentiment analysis of social media posts and found that public sentiment towards data centres has mostly been negative since 2022, with this negative sentiment closely correlated with rising average electricity prices.

In Ireland, electricity from networks built to power new homes has, in various instances, been used up by data centres, while these grid upgrades also led to higher electricity bills for households Hughes (2025). According to a 2026 report commissioned by Beyond Fossil Fuels and Friends of the Earth Ireland, households in Ireland have paid a combined 715 million in electricity costs due to data centres, with such costs likely to remain steep over the coming decade, as reported by Fearon (2026).

This is tied to the question of fairness in cost allocation. According to Peskoe & Martin (2025), the increase in data centre electricity use is pushing utilities to prioritise the needs of “a few energy-intensive consumers”. In some instances, this drives up the general rate for other customers in order to provide discounted rates to data centres, essentially providing “hidden subsidies”.

Satchwell et al. (2025) highlight evolving strategies for how utilities set rates for data centres, in a manner that can give utilities a fair return, reduce the risk of stranded assets, and also protect other customers from an unfair rise in costs. Some notable strategies include making operators sign long-term contracts so that they cannot suddenly exit, or having monthly charges based on peak demand.

The siting of data centres can be based on a diverse range of factors. Network connectivity between different infrastructure sites is one attractive factor, as seen in Northern Virginia, which has a high concentration of data centres in part due to availability of fibre optic loops and cables. Internet connectivity can also help optimise the performance of the data centre according to Nain (2024).

Another factor affecting the location of data centres is the availability of water. The servers in data centres expend a significant amount of waste heat. For cooling, it is common to use a large amount of water. The amount of water consumed by an average data centre in a day could be as much as that consumed by a town of 30,000 to 50,000 people, as per Uddameri (2021). Therefore, contingent upon other factors such as the size of the data centre or technology used, water availability would likely be an important consideration.

The regulatory environment can influence the location of data centres. According to a report by Perner et al. (2025), there are several categories of regulations, such as sustainability, energy sourcing, heat reuse, incentives, and data protection, which can have an impact on either attracting operators or making rules more stringent for them. In addition to this, the Grid connection is also key, as data centres require a vast amount of energy.

Given the complex and vast nature of both the problems and benefits associated with data centres, regulations related to them can take various forms. Below we provide an overview of key regulatory approaches currently in place in Europe, at both a national and EU level, across four categories: energy reporting, sustainability, zoning, and incentives.

Regulation Under the European Union

The regulation of data centres has been part of EU discourse for nearly two decades, as shown in Figure 2.1. In 2008, the Joint Research Commission launched the voluntary EU Code of Conduct on Data Centre Efficiency (European Commission Joint Research Centre 2008), encouraging operators to become "participants" and improve their Data Centre Infrastructure Efficiency, essentially the ratio of power consumed to IT task performed (Bird & Bird LLP 2009). Over the following decade, several indirect frameworks touched on data centre energy use: the Ecodesign Directive (2009/125/EC) set efficiency thresholds for IT hardware (European Union 2009), the Energy Efficiency Framework (2012/27/EU) established broader efficiency targets (European Union 2012), and the Green Public Procurement (GPP) criteria gave preference to more energy-efficient bidders in public IT procurement (European Commission 2020, Council of the European Union 2020).

Directive (EU) 2023/1791, adopted in September 2023, was in particular a watershed moment, as it modified the Energy Efficiency Directive by explicitly introducing regulations related to data centres (European Union 2023). Under Article 12, data centres above a certain size must report a baseline set of indicators, stored in a European database (European Union 2023). Member states transpose directives into national law but may exceed the EU baseline, for instance by lowering reporting thresholds. This means strictness varies considerably across the bloc. The Delegated Regulation (EU) 2024/1364, in force from June 2024, built on this by setting technical details: operators with an installed IT power demand of 500kW or over must report annually on KPIs including energy use, waste heat reuse, and water usage, either through a national reporting system or directly to the European database (European Commission 2024). Being a regulation rather than a directive, it applies to member states directly without requiring transposition.

Energy Reporting

Monitoring and reporting of energy and water-use metrics is one of the most prominent regulatory forms in the EU, aimed at ensuring transparency and accountability. As detailed above, this is centred on the European database for data centres and the KPIs set out under Directive 2023/1791 (European Union 2023) and its 2024 Delegated Regulation (European Commission 2024).

This is echoed and often extended at the national level. France has a transparency framework requiring operators to annually submit reports on indicators like water consumption, waste heat, and IT capacity, published in open data format. Germany's Energy Efficiency Act (EnEfG) goes further still, requiring large data centres to report energy use and efficiency

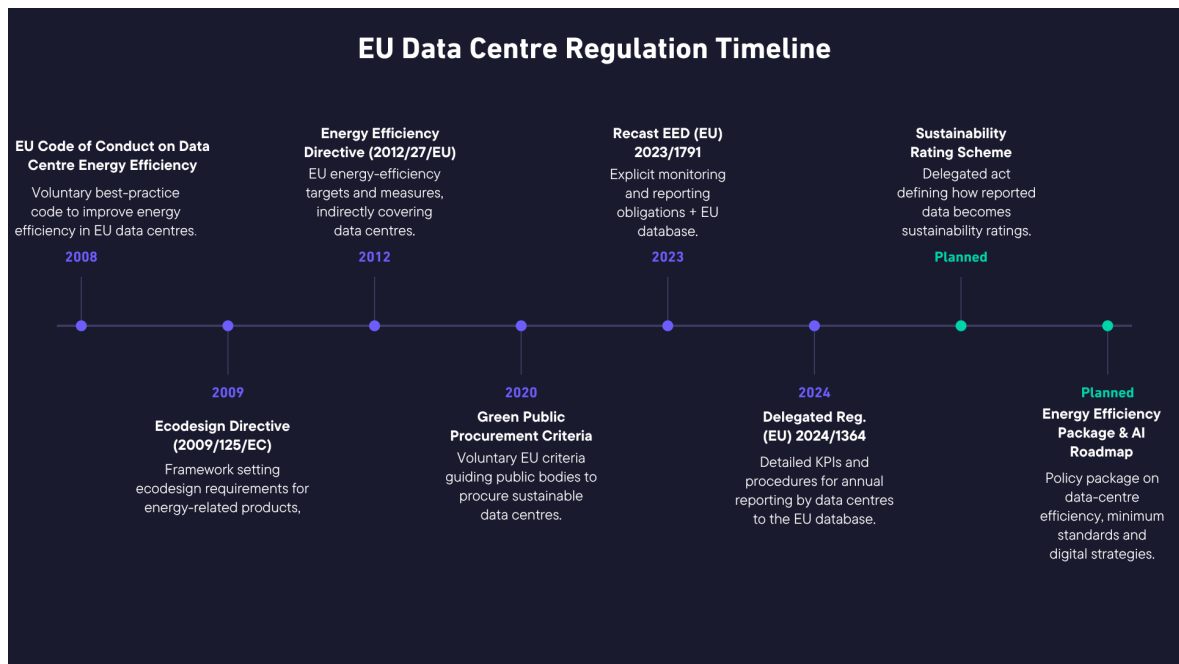


Figure 2.1: Timeline of Key EU Data Centre Regulation

metrics by 31st March each year for publication (Bundesrepublik Deutschland 2023). The Belgian region of Flanders has similar obligations, with reporting due by 15th May. Ireland’s national framework transposes the EU directives, while its Large Energy User Action Plan (LEAP) sets efficiency expectations for big electricity users like data centres (Sustainable Energy Authority of Ireland 2020). Spain is also looking to introduce comparable measures.

Sustainability

Sustainability regulation for data centres broadly takes two forms: waste (or residual) heat reuse requirements, and renewable energy share obligations.

On heat reuse, Article 26(6) of the Energy Efficiency Directive requires member states to ensure waste heat is reused unless operators can prove it is technically infeasible (European Union 2023), while the 2024 Delegated Regulation defined the Energy Reuse Factor to quantify this (European Commission 2024). Germany’s EnEfG sets the strictest national regime, requiring new data centres with a connected load of 300kW or more to hit a minimum energy reuse factor of 10 per cent, rising for facilities built in future years (Bundesrepublik Deutschland 2023). The approach has already shown results elsewhere: waste heat from a 75MW data centre in southern Finland now heats around 2,500 homes in Mäntsälä, cutting residents’ electricity costs (Paulsson et al. 2025), while atNorth’s 30MW SWE02 facility is being built with heat reuse capability to serve local communities in Stockholm (Jackson 2026).

On renewable energy, the Energy Efficiency Directive sets higher-level obligations (European Union 2023) while the Delegated Regulation defines the metrics to measure them (European Commission 2024). Germany again leads here, with the EnEfG requiring data centres to source at least 50 per cent of electricity from renewables from 2024 onwards, rising to 100 per cent from 2027 onwards, but it remains to be seen whether that provision will

actually remain in place (Bundesrepublik Deutschland 2023).

Zoning Rules

Zoning rules govern land use, and in this context determine where data centres can be built and under what conditions. The Netherlands offers the clearest example of restrictive zoning. Its National Strategy on Planning (NOVI) confines large data centres to special business parks where energy demand can be sustainably met, and residual heat can supply district heating (Rijksoverheid 2020), and a 2022 nationwide decision barred hyperscale facilities (over 10 hectares and 70MW) from certain locations (Marel et al. 2022). This is echoed regionally too, with Noord-Holland restricting new data centres to select industrial areas like Amsterdam (Provincie Noord-Holland 2020, Marel et al. 2022), and Haarlemmermeer capping new data centres until 2030, after which no further space will be available (Gemeente Haarlemmermeer 2020).

Elsewhere, France’s Senate bill no. 348 proposes strengthening local authorities’ role in data centre siting alongside environmental safeguards (Sénat de la République française 2023). In Ireland, new data centres require planning permission and are increasingly steered toward locations with spare grid capacity and renewables availability. In Germany, data centres often get placed in commercial and industrial zones, though some regions face environmental permit constraints and municipalities restrict power-intensive facilities in land use plans.

Incentives

Regulation is not always meant to be restrictive. Governments often use policy to attract data centre investment, through mechanisms like tax breaks, discounted energy, streamlined permitting, and guaranteed grid connections.

The UK has a wide range of initiatives in this category. The AI Growth Zones in particular aim to attract large clusters of AI and data centres, through discounted electricity, streamlined planning, and rapid grid expansion. As per the policy paper, this could save up to £80 million annually in electricity bills for a 500MW data centre, according to the now-defunct Department for Science, Innovation and Technology (Department for Science, Innovation and Technology (DSIT) 2025).

France also has compelling incentives for data centres. Operators that commit to a 15 per cent efficiency improvement over five years benefit from a reduced electricity tax (TICFE) (République française n.d.), while EDF’s Project Giga offers land with low-carbon power and efficient grid connections. Greece’s law 4864/2021 provides fast-track permitting and a stable tax regime for projects of ”strategic importance” (Hellenic Republic 2021), Lithuania offers 0 per cent corporate tax and clean energy access in priority areas (Government of Lithuania 2020), and Finland has permitting and electricity benefits for projects that demonstrate efficiency, flexible load, and heat recovery (Prime Minister’s Office, Finland 2025).

3 Methodology

Our methodology is sequenced in four parts which, considered together, inform our final insights.

We begin with a comprehensive analysis of regulatory frameworks’ impact on data centres across European and Eurasian countries. We hand-encode four different regulatory categories as factor variables: energy reporting or monitoring obligations for data centres, sustainability obligations on operators, planning-related constraints, and incentives to attract operators. We then conduct exploratory analysis of the relationship between data centre presence and state-level energy prices (our two key variables of interest) across all 51 US states using regression. Following this US-based exploration, we join our hand-coded regulation variables with existing data from ENTSO-E (2026) and Ember (2026) including international energy trade, energy generation mix, and population, to form a master dataset comprising 30 European countries (see Section 3.3.3). Several existing frameworks served as sources of inspiration for our covariate set. The Energy Efficiency Directive, legislated by the European Commission, as well as a white paper authored by Lin & Bunger (2023) on behalf of Schneider Electric, are crucial among these. Lin identifies several key metrics: energy, greenhouse gas emissions, water, waste, and local ecosystem. These, along with data availability from ENTSO-E (2026) and Ember (2026), laid the path that guided the building and cleaning of our dataset. Final data covers the years 2023–2025, indexed by Country-Year.

Our statistical analyses of this data begin with Energy Flow Network Modeling, which aims to uncover the impact of the energy trade network structure on data centre presence and electricity prices by studying a network of European countries linked by cross-border electricity trade. We employ three complementary methods, starting with Exponential Random Graph Models (ERGMs) to test whether our key variables associate with the presence of energy trade ties. Then, Network-Exposure Cross-Sectional Regressions explore these effects further, followed by Network Autocorrelation Models to directly regress these outcomes on covariates while accounting for interdependence among trading partners and test for country interdependence.

Causal analysis builds on this, leveraging key developments in the field which have enabled an efficient and streamline process to identify, estimate, and refute causal effects between treatments and outcomes. Literature suggests that the rapid growth of data centres is promoted by factors such as energy prices, fuel for energy generation, and beneficial regulatory frameworks. This method hinges on recently developed frameworks that streamline causal analyses; the DoWhy library proposed by Sharma & Kiciman (2020) and the DoWhy-GCM extension by Blöbaum et al. (2024).

Each of these methods, in that order, are outlined along with results in this chapter.

3.1 Regulation Encoding

We assess a set of 45 European and Eurasian countries based on the data centre regulations highlighted in the previous section of this report. The different regulation types were each treated as factor variables, hand-encoded according to the strength or strictness of the regulation. We manually read over 60 relevant policies and directives across countries, before once again manually applying our encoding criteria, and verifying using Claude Sonnet 5 (Anthropic 2026), which flagged no concerns. First, we one-hot encoded each country based on whether it is currently part of the European Union. In the `in_eu` column, countries that are officially part of the European Union were labelled as 1. On the other hand, countries that are not official members of the EU, regardless of whether they have previously or currently sought candidacy, were labelled as 0. We then label-encoded each country on a scale of 0 to 3 for each of the following regulatory aspects (one column per dimension): energy reporting, sustainability, zoning, incentives, and attractiveness. The scale is ordinal, with 0 broadly representing no meaningful provisions and 3 representing strong and stringent national frameworks.

Reporting. This column measures whether a country has binding provisions that require data centres to monitor and report various indicators such as energy use, energy effectiveness, or water use. We assess if this reporting process is mandatory, has set standardised procedures, and gets directed to an official database. The scoring is as follows:

- 0** There are no regulations specifically requiring data centre operators to report energy-related performance indicators.
- 1** Baseline reporting without substantial national-level additions. This includes countries that need to follow Article 12 of the EU Energy Efficiency Directive and the Delegated Regulation 2024/1364, but have no additional national-level regulations.
- 2** Moderate national-level obligations, beyond just the baseline obligations set out by organisations such as the EU.
- 3** Strong national-level data centre-specific guidelines, with energy thresholds lower than the baseline and stricter enforcement processes.

Sustainability. This column follows a similar set of criteria to answer the question of whether there are binding provisions which require data centres to meet targets related to waste-heat reuse (i.e. a particular percentage of waste heat has to be reused) or some sustainability-related thresholds (i.e. a certain percentage of the energy use has to be powered by renewable energy sources). The scoring was done as follows:

- 0** There is no data centre-specific sustainability framework, and no laws that set specific residual heat or renewable share obligations.
- 1** Baseline sustainability expectations, based on legislation by multilateral organisations, but no additional national-level quantitative targets.

- 2 Moderate national-level sustainability requirements, clearly addressing data centre-type infrastructure and going beyond the baseline. Includes various grid-linked renewable requirements or mandatory residual heat use assessments.
- 3 Strong national-level sustainability regimes, referring to countries that impose explicit and binding targets such as minimum heat reuse thresholds, progressively stricter phase-in requirements, and specific dates for renewable energy targets.

Planning. This column measures the extent to which general zoning, planning, and environmental assessment rules restrict the locations where infrastructure such as data centres can be built. It also considers whether there are explicit bans or dedicated categories for such entities. A higher score in this category implies the imposition of additional layers of constraints beyond standard commercial projects.

- 0 Minimal constraints; data centres and similar infrastructure can be built almost anywhere.
- 1 Standard planning rules where data centres are treated like generic industrial or commercial cases, and have to comply with general zoning/permits.
- 2 Countries with clear additional constraints, where grid or environmental rules materially constrain siting for data centres. Includes places with strong preconditions for consent related to grid capacity/environmental protection.
- 3 Countries with data centre-specific zoning bans or tight national or regional restrictions which limit new large data centres to specifically designated zones. New large data centres are often restricted to a few areas that can sufficiently match the grid energy requirements.

Incentives. This column measures the overall incentives and attractiveness of data centres and related investment across different national tech ecosystems. Specifically, we look at a variety of formal policy instruments that reduce cost for such infrastructure. This includes tax regimes, whether corporate taxes are low, whether lucrative exemptions are in place, and if there are any credit systems for Research and Development (R&D). This column required slightly more subjective discretion to quantify, as it incorporates several components.

- 0 Very weak attractiveness; no incentives in place.
- 1 General incentives for investment and technology, meaning moderate ease of doing business, but no significant measures targeting data centre infrastructure.
- 2 Clear advantages for data centres and similar tech infrastructure, including Special Economic Zones with tax breaks/VAT exemptions, low de facto corporate tax rates, and targeted support for digital infrastructure.
- 3 Very strong system of incentives targeting data-related tech, including energy price support, AI growth zones, and national consent regimes.

3.2 Exploration of US Data Centres and Energy Prices

As an initial step, we ran a cross-sectional analysis looking at the relationship between data centre presence (measured both as raw count and as density per capita) and state-level energy prices across all 51 US states. This was controlled for population, GDP per capita, and corporate tax rates, since these are all factors that could plausibly affect where data centres end up as well. The idea here was not to draw firm conclusions, but rather to get an initial sense of whether there is any visible relationship worth digging into further.

We pulled data centre counts per state from (Data Center Map 2024), which gave us a total of 3,996 operational facilities across all 51 states. Commercial and industrial electricity prices by state came from the Electric Power Monthly Table 5.6.A (November 2025) from (U.S. Energy Information Administration 2026). State population and geographic area figures were taken from the US Census Bureau’s 2020 Decennial Census (U.S. Census Bureau 2020), which we used to calculate data centre density (data centres per million population) as well as population density more broadly. GDP per capita by state came from (U.S. Bureau of Economic Analysis 2024), and corporate income tax rates came from (Tax Foundation 2024), which range from zero (six states with no corporate tax) up to 11.5 per cent in New Jersey.

Looking at the raw correlations, the relationships between data centre count and energy prices were generally weak. Data centre count against commercial energy price came out at a weak negative correlation of -0.160 , which was not significant at the 5% level ($p = 0.2625$). Against industrial energy price it was weaker still, at -0.110 ($p = 0.4434$), and against corporate tax rate it was -0.146 ($p = 0.3056$), though it is worth noting that Texas and Ohio, which appear to benefit from having no corporate tax rate, actually rely on a gross receipts tax instead.

By far the strongest relationship in this initial look was between data centre count and state population, at 0.653 and highly significant ($p < 0.001$). State area on its own showed only a weak positive correlation of 0.114 , again not significant ($p = 0.4255$). A few states stood out as outliers throughout. Virginia had a very high data centre count alongside only moderate prices, Texas combined low prices with an extremely high data centre count, and California had both high prices and a large number of data centres.

Once we normalised for population and looked at data centre density (data centres per million people) rather than raw count, the picture shifted a bit. The correlation with commercial energy price strengthened to -0.350 , and this time it was significant at the 5% level ($p = 0.0119$), so it seemed that lower energy prices might align more closely with higher data centre density once differences in state size were accounted for.

A regression controlling for GDP per capita and corporate tax rate (with Virginia excluded as an extreme outlier) suggested commercial energy prices were a significant factor ($p = 0.014$), with each 1¢/kWh increase associated with a 0.54 reduction in data centre density per million people, though the model only explained about 21% of the variance ($R^2 = 0.213$) and neither GDP per capita nor corporate tax rate came out as significant. We want to be clear that this is only an association picked up in an exploratory look at the data, not a causal claim. Figure 3.2 shows the regression, with the actual and predicted data centre density (by population).

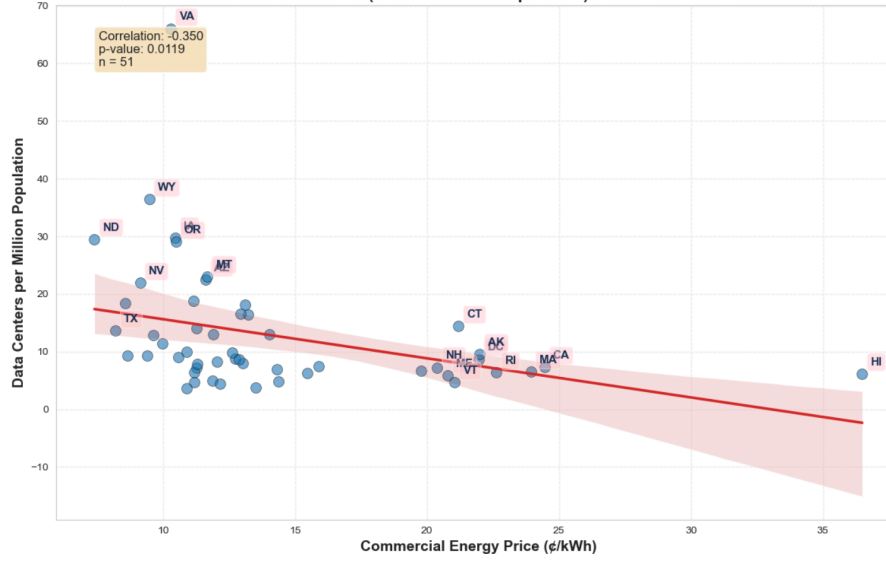


Figure 3.1: Data Centre Density vs Commercial Energy Price

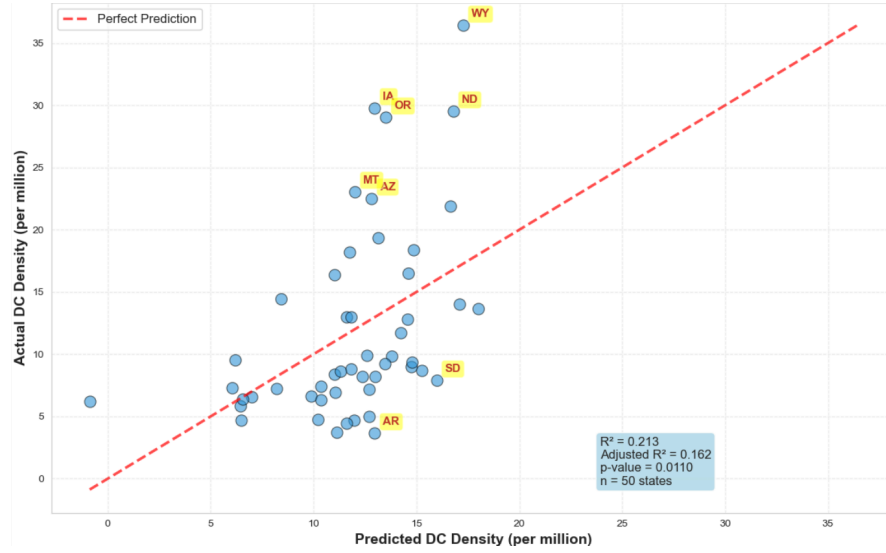


Figure 3.2: Regression Analysis of US Data Centre Presence

Virginia in particular seemed to be its own story rather than something these variables could really explain (evident by Virginia's clear outlier position in the top-left corner of Figure 3.1); reporting has pointed to developers recognising the area as ideal for data centres due to flat, cheap land (Faguy 2025), a long history of network infrastructure investment dating back to the 1960s, and proximity to both the Pentagon and several universities, providing a mix of talent and demand. It should be noted, however, that wholesale electricity costs in areas near data centres have reportedly risen by as much as 267% (Saul et al. 2025), which hints that state-level averages may be masking more significant local price effects.

This US analysis provides a foundation for our study as a whole, having explored the relationship between our two key variables. However, data availability for regulatory frameworks and energy-related variables was more abundant and accessible in European regions, so the remainder of the analysis focuses on Europe.

3.3 Energy Flow Network Modeling

This section addresses the methodological question of whether structural features of the European cross-border electricity trade network are associated with two outcomes of substantive interest, **wholesale electricity price** and **data centre presence**. Rather than a single model, this is addressed through a three-stage pipeline, each stage answering a distinct question and directly informing the specification of the stage that follows. This structure is summarised in Figure 3.4 before each stage is developed in full in Sections 3.3.4–3.3.6. Figure 3.3 shows the exact networks in question.

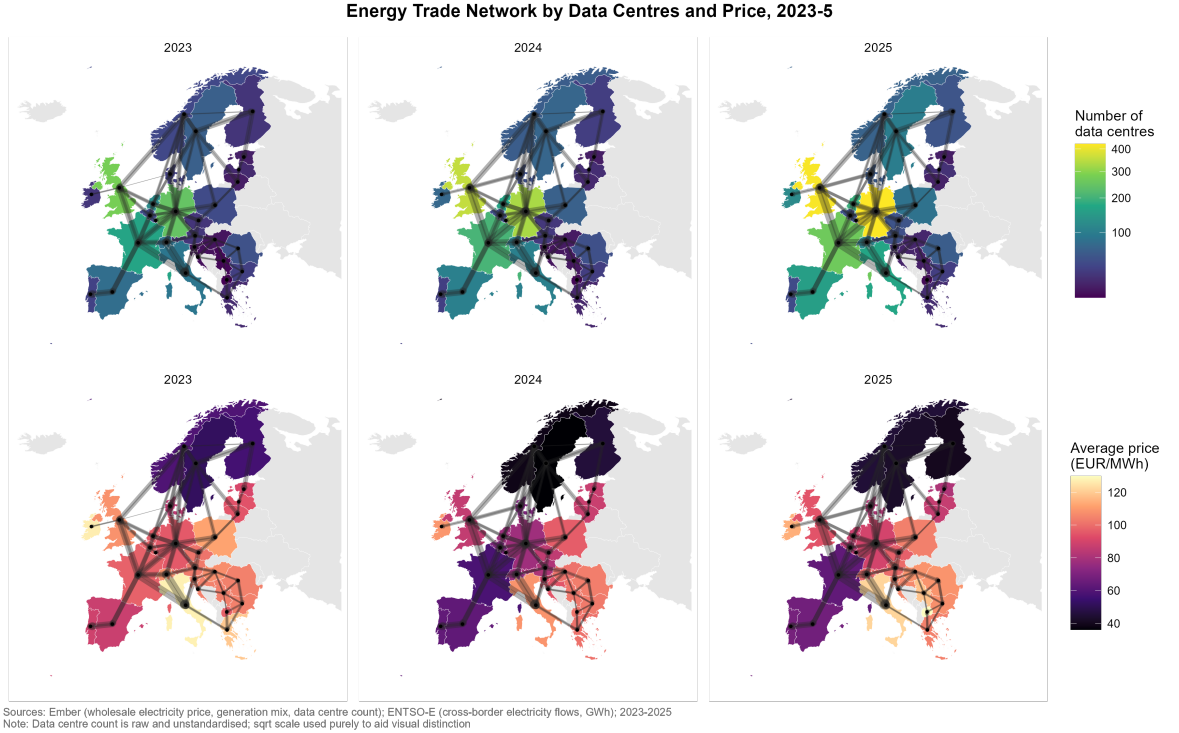


Figure 3.3: Illustrative visualisations of the European Energy Trade Network for Jan-Sep 2025, overlaid onto geographical heatmaps showing our two key variables: data centre count and energy price.

3.3.1 Overview of the Modelling Workflow

The pipeline begins with an **Exponential Random Graph Model (ERGM)**, fit to the observed trade network itself (Section 3.3.4). Here, the outcome being modelled is the network’s structure, asking which pairs of countries trade electricity with one another. This stage identifies which country-level attributes, and which structural properties of the network, are associated with the presence of a trading relationship. Its role in the wider pipeline is diagnostic, establishing *which* variables show a genuine relationship with network position, and *how* (e.g. whether it is a country’s price level, or the price gap between trading partners, that matters), before those variables are carried forward into models of the two outcomes of interest.

The second stage translates these tie-level (dyadic) findings into **country-level cross-**

sectional regressions (Section 3.3.5), with data centre presence and electricity price as the outcomes. Because the ERGM’s findings concern relationships *between* pairs of countries, they cannot be entered into a country-level regression directly; instead, network-exposure variables are constructed (for example, a country’s average price gap with its trading partners) that translate a dyadic finding into a property of each individual country. This stage tests whether the structural relationships identified by the ERGM extend beyond predicting tie formation to explaining the two outcomes this analysis is ultimately concerned with.

The third stage, a **Network Autocorrelation (Spatial Autoregressive) Model** (Section 3.3.6), formalises the network-exposure approach used in the second stage. Rather than treating a country’s exposure to its neighbours as an ordinary, pre-computed regressor, this model estimates network dependence directly and jointly with the rest of the specification, correcting for a statistical issue (reflexivity, explained fully in Section 3.3.6) that the second stage’s approach cannot address. This stage is used both as a robustness check on the cross-sectional findings and as the more rigorous, defensible test of whether network position itself, rather than an engineered proxy for it, explains variation in the two outcomes of interest.

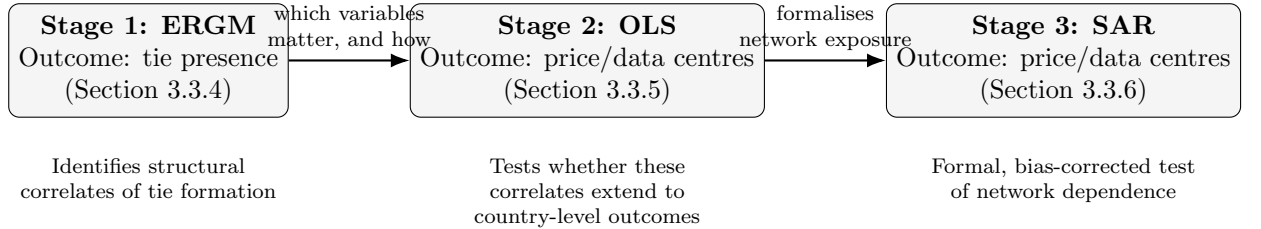


Figure 3.4: Overview of the three-stage modelling pipeline. Each stage answers a distinct question and directly motivates the specification of the next.

3.3.2 A Concrete Illustration of How the Pipeline is Used

The ERGM (Section 3.3.4) finds that the *difference* in wholesale price between two countries, not either country’s level in isolation, robustly predicts tie formation. This gives insights about network structure, not directly about price or data centre presence. The second stage (Section 3.3.5) operationalises this by constructing each country’s average price gap with its trading partners and testing its association with data centre presence. The third stage (Section 3.3.6) asks a related question of the same relationship, seeking to test whether a country’s own price moves with its partners’ prices once network dependence is modelled formally, rather than approximated by an exposure variable. Each stage thus builds on the substantive content of the last, rather than repeating three independent analyses of the same data.

All three stages operate on the same annual, undirected trade network (2023-2025) and country-level attributes, constructed and justified in full in Section 3.3.3.

3.3.3 Data and Pre-Processing

All three modelling stages operate on a single, common set of pre-processed data. Pre-processing comprises network construction, country and feature selection, conversion to

undirected edges, network structural stability checks, skewness checks for continuous variables, correlation-driven feature engineering, thin-category handling and confirmation of time-invariance. Full source code (in *R*) for every step is available in the project repository.

Cross-border physical electricity flows are drawn from ENTSO-E (2026)’s Physical Energy & Power Flows dataset (2023–2025, monthly resolution per ordered country pair); country-level attributes (price, generation mix, demand) are drawn from Ember (2026), data centre counts from Data Center Map (2024), population and land area from World Bank (2026*b*) and World Bank (2026*a*) respectively, EU membership was hand-encoded and the four regulatory indices were hand-encoded following direct review of national regulatory frameworks (Section 2).

Annual edge lists sum monthly export flows (GWh) per country pair to annual totals. Prior to further pre-processing, country names were standardised before harmonising codes to ISO3 via a lookup table (IBAN 2026), with an explicit check for unmatched codes.

Countries were retained only with complete attribute data across all three years; of 30 countries covered, 1 (Montenegro) was excluded for incomplete year coverage, leaving 29 countries. This filtering was applied identically to edge lists and attribute tables, ensuring an identical node set across all three annual networks.

The attribute dataset initially included a substantially larger candidate variable set. A correlation/Variance Inflation Factor (VIF) screen removed exact or near-exact linear functions of other retained variables. ($VIF_k = 1/(1 - R_k^2)$, where R_k^2 is the R^2 from regressing predictor k on every other predictor.) These were `pct_clean` and `pct_fossil` (exact linear combinations of `pct_renewable` and `pct_nuclear`, the three shares summing to 100%); `demand_twh`, `total_gen_twh`, and `net_imports_twh` (related by an accounting identity and each correlated with population at $r = 0.93$ – 0.99 , indicating they are primarily size proxies); and `co2_intensity_g_kwh` (redundant with retained generation-mix variables, $r = 0.92$).

The retained set comprises `price_eur_mwh` and `data_centers` (both variables and candidate outcomes); `pct_renewable`, `pct_nuclear`, `pct_hydro`; `demand_per_capita`; `population`, `land_area_sqkm`, `in_eu`; and the four regulation indices, kept separate despite `sustainability` and `reporting`’s correlation ($r = 0.89$) given regulation’s status as a primary research focus. This decision was later revisited after correlation checks, which are explained shortly.

Reciprocity, defined here in Equation 3.1 at the level of tie existence rather than flow volume, measures the number of dyads for which a directed edge exists in both directions, as a proportion of the number of dyads for which a directed edge exists in at least one direction.

$$\text{Reciprocity} = \frac{|\{(i, j) : i \rightarrow j \text{ and } j \rightarrow i\}|}{|\{(i, j) : i \rightarrow j\}|}, \quad (3.1)$$

This was checked directly and found to be 0.99–1.00 in every year. A directed specification is only meaningful where genuine directional asymmetry exists, and is otherwise unstable to fit. The network was therefore collapsed to undirected. Edge weights were constructed by summing, rather than taking the net of, the flow between connected dyads in both directions. This choice reflects the focus of this analysis on whether network *connectivity* and trade *intensity* relate to data centre presence and price. Net flow would be a poor measure of connectivity in this particular network. Reciprocity of 0.99–1.00 (Section 3.3.3) confirms

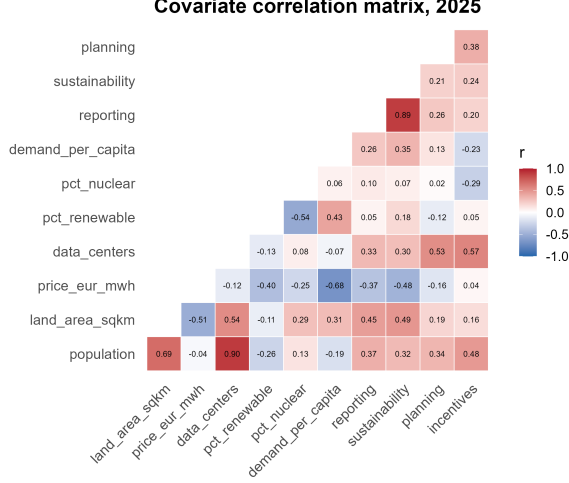


Figure 3.5: Covariate Correlation Heatmap

that positive flow exists in both directions for nearly every trading pair, but reveals nothing about the relative magnitude of those flows. Directly checking the ratio of larger to smaller directional flow for each reciprocal pair shows a median imbalance of $6.2\times$ (mean $54.7\times$, driven by several pairs exceeding $100\times$). Given this pronounced directional asymmetry, net flow would substantially distort the resulting edge weights. Furthermore, the minimum is a $1.04\times$ ratio, which would reduce a substantial, large-volume trading relationship to a value near zero. Summing avoids this distortion, preserving the full bilateral trade volume as the edge weight regardless of its directional balance. The resulting undirected edge count (59, identical in all three years) is consistent with the directed counts (117 in 2023/2024, 118 in 2025).

The edge set was found to be identical at the binary level across all three years (59/59 shared, Jaccard = 1.000 throughout), while data centre count varied substantially over the same period (mean absolute year-on-year change of 27.3% and 36.3%). This establishes that variation in later results reflects changing country attributes, not a changing network; the empirical basis for the time-invariance discussion below.

Every continuous covariate was screened for extreme skew (max:min ratio). Population ($121\text{--}125\times$) caused non-convergence in preliminary ERGMs under every continuous transformation tried, and was discretised into terciles (small/medium/large) for use as a nodal factor. Land area ($209\times$, unchanged year-to-year) received identical treatment, as a second size-control option reserved for a final robustness check (Section 3.3.4.7). data centre count ($55\text{--}60\times$) was retained continuous.

A full covariate correlation check, shown in Figure 3.5 identified two pairs warranting a constructed alternative (**data_centers** $\times$ **population**, $r = 0.897$; **reporting** $\times$ **sustainability**, $r = 0.887$), retaining the originals alongside each construction for later robustness checks.

A per-capita data centre variable was constructed as raw data centre count divided by $\text{population} \times 10^6$, reducing the skew ratio to $13\text{--}25\times$ and its correlation with population to -0.317 . A composite regulation variable, **sustainability_reporting**, was constructed as the rounded mean of its two correlated parents. Being an almost deterministic function of

them ($r = 0.93\text{--}0.97$), it was never used alongside either parent, but these alternative options were used in robustness checks (Table 3.1).

This yields three deliberate parallel-specification axes (Table 3.1), each tested explicitly as a robustness check in the sections that follow by running two almost identical models, but switching option A for B in each and assessing whether effects survived or emerged.

Robustness axis	Option A (primary)	Option B (comparison)
Country size control	Population	Land area
data centre measure	Raw count	Per-capita
Regulation (reporting/sustainability)	Kept separate	Combined composite

Table 3.1: The three parallel-specification axes built into pre-processing, each carried forward as an explicit robustness check.

The four regulation indices share a common intended 0–3 ordinal scale; factor levels were set explicitly to $\{0, 1, 2, 3\}$ rather than inferred per year, so the reference category and dummy structure stay identical regardless of which cells are populated in a given year. Categories with very few observations can produce unstable or non-identifiable estimates, so some were merged. A threshold of 3 was adopted to flag thin categories. The possibility for a higher threshold was considered, but given our small sample size of $n = 29$, this would force aggressive collapsing across nearly every regulation variable across up to four categories, destroying much of the hand-coding’s granularity. In our sample, three is the minimum sufficient to rule out a single country deterministically fixing a category’s effect, while preserving most of the original structure. Every flagged category shown in Table 3.2 is revisited with an explicit exclusion-sensitivity test wherever it produces a reportable result (Sections 3.3.4.7, 3.3.5.7).

Variable	Category	n countries	Countries
reporting	3	1	Germany
sustainability	3	1	Germany
sustainability_reporting	3	1	Germany
planning	3	1	Netherlands
incentives	3	2	United Kingdom, Ireland

Table 3.2: Regulation categories below the minimum cell-size threshold (3 countries), flagged for exclusion-sensitivity testing. Identical across all three years (2023–2025).

Population, though technically time-varying, moves by a mean of only 0.57% per year, which is too little to shift any country across a tercile boundary, confirmed directly via the identical `pop_tercile/land_area_tercile` values in every year. Combined with the regulation indices (fixed by construction) and EU membership (unchanged), every control variable is time-invariant across 2023–2025; only `price_eur_mwh` and `data_centers` vary meaningfully. Consequently, any difference in a model’s coefficients across years reflects a genuine change in the underlying relationship or in these two variables specifically, not drift in any control.

3.3.4 Exponential Random Graph Models

3.3.4.1 Motivation and Novelty

Exponential Random Graph Models (ERGMs) provide a principled way to model the joint probability distribution of an entire observed network, rather than treating each potential tie as an independent binary outcome (Robins et al. 2007). Reciprocity in the observed trade network was found to be 0.99–1.00 in every year (Section 3.3.3), meaning tie formation cannot be adequately explained by treating each potential connection between two countries as statistically independent of every other connection in the network. In other words, whether countries i and j trade plausibly relates to the broader structure of the network, not on i and j ’s attributes alone. ERGMs address this directly by modelling the full joint distribution of the network via an exponential-family likelihood defined over network-level statistics (formalised in Equation 3.2), rather than via a sequence of independent per-dyad probabilities. Our hypothesis is that our key variables of interest, data centre presence and energy prices, may have some association with the energy trade network structure, in either direction. This motivates our exploration of the energy network in relation to these variables.

ERGMs have previously been applied to power grid structure by (Giacomarra et al. 2023), in the context of generating synthetic power grids. That study’s methodology is geography-agnostic and its empirical validation relies primarily on synthetic and benchmark grids (most centrally the IEEE 300-bus network, a synthetic academic benchmark). This type of modelling originated in sociology and statistics, developed to model the formation of social ties on the premise that individuals cannot be treated as independent of their social context (Holland & Leinhardt 1981). We hypothesise that treating countries in a connected energy flow network as independent from their trade partners would be similarly misguided. Since energy prices and data centre presence are entangled in this environment, this is a key consideration. To the best of our knowledge, and following a dedicated literature search, no prior study applies ERGMs to a real-world, observed international electricity trade network of the kind analysed here; this analysis therefore represents a novel empirical application of the ERGM framework, distinct in both data and purpose from existing work in this area.

3.3.4.2 Model Formulation

An ERGM specifies the probability of observing a particular network y , out of the set $\mathcal{Y}$ of all possible networks on the same set of nodes, as:

$$P_{\boldsymbol{\theta}}(Y = y) = \frac{\exp(\boldsymbol{\theta}^\top \mathbf{g}(y))}{\kappa(\boldsymbol{\theta})}, \quad y \in \mathcal{Y} \quad (3.2)$$

where Y is the random network, y a particular realisation of it (here, the observed 2023, 2024, or 2025 trade network); $\mathbf{g}(y)$ is a vector of *network statistics*, which are summaries of y chosen by the analyst, such as the total number of ties, or a measure of how a country’s price level relates to its ties (defined term-by-term below); $\boldsymbol{\theta}$ is the corresponding vector of parameters to be estimated, one per statistic in $\mathbf{g}(y)$; and $\kappa(\boldsymbol{\theta}) = \sum_{y' \in \mathcal{Y}} \exp(\boldsymbol{\theta}^\top \mathbf{g}(y'))$ is a normalising constant, summing over every possible network on the same node set so that

Equation 3.2 is a valid probability distribution.

For a network of even modest size, $\mathcal{Y}$ is astronomically large (for $n = 29$ undirected nodes, $|\mathcal{Y}| = 2^{\binom{29}{2}} = 2^{406}$), so $\kappa(\boldsymbol{\theta})$ cannot be computed directly, and Equation 3.2 cannot be maximised by conventional means. This intractability is the central estimation challenge that Algorithm 3.6 is designed to address.

3.3.4.3 Interpreting $\boldsymbol{\theta}$ the Conditional Log-Odds of a Tie

Although $\kappa(\boldsymbol{\theta})$ cannot be computed, Equation 3.2 implies a tractable and directly interpretable expression for the probability of a single tie, conditional on every other tie in the network being held fixed:

$$\text{logit}[P_{\boldsymbol{\theta}}(Y_{ij} = 1 \mid Y_{-ij} = y_{-ij})] = \boldsymbol{\theta}^\top \boldsymbol{\delta}_{ij}(y) \quad (3.3)$$

where Y_{ij} is the (random) tie between nodes i and j , Y_{-ij} denotes every other tie in the network, and $\boldsymbol{\delta}_{ij}(y) = \mathbf{g}(y_{ij}^+) - \mathbf{g}(y_{ij}^-)$ is the *change statistic*, i.e. the difference in the network statistics $\mathbf{g}(\cdot)$ between the network with the ij tie present (y_{ij}^+) and absent (y_{ij}^-), holding every other tie fixed. A positive θ_k means that, holding the rest of the network fixed, a change that increases the k -th network statistic (e.g. two countries having a smaller price gap) increases the conditional log-odds of a tie existing between them.

3.3.4.4 Estimation Algorithm (MCMC Maximum Likelihood)

Because $\kappa(\boldsymbol{\theta})$ is intractable, $\boldsymbol{\theta}$ cannot be obtained by direct likelihood maximisation. Where the specified statistics $\mathbf{g}(y)$ are *dyad-independent* (each term is a function of individual ties only, with no dependence between them. This is true of every term in this analysis except the transitivity term introduced last, the pseudo-likelihood exactly equals the true likelihood, and $\boldsymbol{\theta}$ can be obtained directly via logistic regression on every dyad. Once a *dyad-dependent* term (transitivity) is introduced, this shortcut no longer holds, and the model is instead fitted by Monte Carlo Markov Chain Maximum Likelihood Estimation (MCMC-MLE), following (Hunter et al. 2008, Geyer & Thompson 1992), summarised in Algorithm 3.6.

Every specification in this analysis was fitted in R using the `ergm` package (Hunter et al. 2008). Terms that are dyad-independent (Steps 2-4 collapse to Step 1 exactly, with no MCMC sampling required) were fitted first, in stages, and the dyad-dependent transitivity terms were added only once every nodal-covariate term had been introduced, so that any convergence problem could be attributed unambiguously to the newly-added term rather than an unknown interaction between several simultaneously-introduced terms.

3.3.4.5 Terms Used in the Final Specification

As the network is undirected (Section 3.3.3), sender/receiver effects and a reciprocity term (`nodeicov/nodeocov/mutual`, used in a directed specification) collapse into a single tie-presence outcome and a single nodal-covariate effect per variable (`nodecov`). Each right-hand-side term below corresponds to a specific choice of $\mathbf{g}(y)$, motivated by the substantive question it answers. For a nodal covariate x :

Algorithm: MCMC Maximum Likelihood Estimation for ERGMs

1. **Initialise** $\theta^{(0)}$ via Maximum Pseudo-Likelihood Estimation (MPLE): logistic regression of tie presence on the change statistics $\delta_{ij}(y)$, treating dyads as independent.
2. **Simulate** networks $\{y^{(1)}, \dots, y^{(m)}\}$ from the current estimate $\theta^{(t)}$ via Metropolis–Hastings: propose toggling a random dyad, accept/reject based on the resulting change in $\theta^{(t)\top} \delta_{ij}(y)$ (Equation 3.3).
3. **Approximate** the log-likelihood ratio between a trial θ and $\theta^{(t)}$ via importance sampling over the simulated networks, avoiding direct computation of $\kappa(\theta)$.
4. **Update** $\theta^{(t+1)}$ by maximising this approximation (Newton–Raphson), matching simulated statistics to the observed $\mathbf{g}(y_{\text{obs}})$.
5. **Iterate** Steps 2–4 until the change in θ is negligible relative to its standard error.
6. **Diagnose**: check MCMC trace plots for stationarity, and assess goodness-of-fit against the observed network’s structural properties.

Figure 3.6: MCMC-MLE estimation procedure for ERGMs, following (Geyer & Thompson 1992), as implemented in the `ergm` package (Hunter et al. 2008).

$$g_{\text{nodecov}(x)}(y) = \sum_{(i,j) \in E(y)} (x_i + x_j) \quad (3.4)$$

$$g_{\text{absdiff}(x)}(y) = \sum_{(i,j) \in E(y)} |x_i - x_j| \quad (3.5)$$

$$g_{\text{nodematch}(x)}(y) = \sum_{(i,j) \in E(y)} \mathbb{I}(x_i = x_j) \quad (3.6)$$

where $E(y)$ denotes the set of ties present in network y . Equation 3.4 (`nodecov`) tests whether a country’s level of x (e.g. price) is associated with tie formation as a general, additive main effect; a high value of x raises the log-odds of a tie regardless of which specific partner is involved. Equation 3.5 (`absdiff`) instead tests whether the *difference* in x between two countries predicts a tie between them, a genuinely relational quantity depending on both countries’ values jointly, rather than either in isolation. Equation 3.5 (`absdiff`) tests whether the *difference* in x between two countries predicts a tie between them, independent of either country’s individual level; Equation 3.6 (`nodematch`) tests whether two countries sharing the same categorical value of x (here, EU membership) are more likely to trade. For a categorical nodal covariate with K levels, `nodefactor` enters $K-1$ indicator terms (one per non-reference category), each following the same form as Equation 3.4 with x replaced by the corresponding category indicator.

Finally, transitivity is captured via the geometrically weighted edgewise shared partners statistic (Hunter et al. 2008, Snijders et al. 2006):

$$g_{\text{gwesp}(\alpha)}(y) = e^\alpha \sum_{k=1}^{n-2} \left\{ 1 - (1 - e^{-\alpha})^k \right\} EP_k(y) \quad (3.7)$$

where $EP_k(y)$ is the number of pairs of connected nodes sharing exactly k common partners, and α (fixed at 0.25 throughout, following standard practice) is a decay parameter controlling how quickly the contribution of additional shared partners diminishes, a construction specifically designed to avoid the well-documented model degeneracy that a raw triangle count is prone to (Snijders et al. 2006). This term tests whether two countries sharing a common trading partner are more likely to trade with one another directly than the nodal covariates alone would predict. This is a structural, network-endogenous effect, distinct from every other term in the specification, and therefore introduced only once every nodal covariate had already been included, so that its estimated effect is not confounded with an unmodelled nodal relationship that happens to resemble clustering.

The following *R* code reproduces the fitting of the final specification (2025, raw data centre count):

```
m6_raw_2025 <- ergm(
  net_2025 ~ edges +
    nodecov("price_eur_mwh") + absdiff("price_eur_mwh") +
    nodecov("data_centers") + absdiff("data_centers") +
    nodefactor("pop_tercile") +
    nodecov("pct_renewable") + nodecov("pct_nuclear") +
    nodematch("in_eu") +
    nodefactor("sustainability_reporting") +
    nodefactor("planning") + nodefactor("incentives") +
    gwesp(0.25, fixed = TRUE),
  control = control.ergm(MCMLE.maxit = 40))
```

3.3.4.6 Key Results (2025, Final Specification)

Table 3.3 reports the fully-specified model for 2025, after every addition in the build described above (baseline; price and data centre count; country size; generation mix; regulation; transitivity). Equivalent specifications were fitted independently for 2023 and 2024 (available in the project repository), each converging without degeneracy, and each with directly comparable estimated effects for the two variables of central interest.

Two findings are robust across all three years and both parallel key-variable specifications (raw and per-capita data centre count), and form the core structural results of this analysis. First, the price gap between two countries (Equation 3.5) is a consistently significant, negative predictor of tie formation in every year ($p < 0.01$ throughout). We find that countries with more similar wholesale prices are more likely to trade, a result running counter to a naive arbitrage account, in which larger price gaps would be expected to *increase* the incentive to trade. Second, transitivity (Equation 3.7) is significant in every specification ($p < 0.001$, coefficient stable at 1.09–1.14), confirming genuine network-structural clustering beyond what any nodal covariate explains.

Term	Estimate	Std. Error	<i>p</i>
Edges	−0.232	2.458	0.925
Price level (<code>nodecov</code>)	−0.008	0.007	0.257
Price gap (<code>absdiff</code>)	−0.047	0.011	< 0.001 ***
Data centres, level (<code>nodecov</code>)	0.012	0.004	0.001 **
Data centres, gap (<code>absdiff</code>)	−0.005	0.002	0.036 *
Population tercile: medium	0.680	0.398	0.088
Population tercile: small	0.943	0.477	0.048 *
% Renewable generation	−0.016	0.009	0.069
% Nuclear generation	−0.016	0.010	0.104
Shared EU membership	−0.339	0.538	0.528
Regulation (sustainability/reporting): levels 1–3	—	—	all <i>p</i> > 0.36
Planning: level 2	−0.526	0.270	0.051
Planning: level 3	−0.716	0.560	0.202
Incentives: level 2	−0.593	0.319	0.063
Incentives: level 3	−2.127	0.960	0.027 *
Transitivity (<code>gwesp</code> , $\alpha = 0.25$)	1.090	0.312	< 0.001 ***

Table 3.3: Final ERGM specification, 2025 (raw data centre count). AIC = 294.4, BIC = 366.5. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

data centre count (Equation 3.4) is significant and positive in every year of the raw specification, but (as detailed in the robustness section below) this finding does not survive a change in the choice of country-size control, and should be interpreted with this caveat attached rather than as an unconditionally established result.

3.3.4.7 Robustness Checks

Three robustness checks were carried out, each targeting a specific, pre-identified concern rather than applied as a blanket procedure to every coefficient in the model.

Sensitivity to extreme observations. Both `data_centers` and its per-capita counterpart are heavily right-skewed (Section 3.3.3). For `dc_per_capita`, the distribution of the country most and second-most exposed to this skew (Luxembourg and Ireland, consistently across years) was examined directly, and the price-gap-equivalent `absdiff(dc_per_capita)` term was refitted excluding these countries. The effect strengthened, rather than weakened, upon exclusion (e.g. 2024: -0.120 , $p = 0.042$ in the full sample versus -0.172 , $p = 0.021$ with Luxembourg and Ireland excluded), indicating the finding is not an artefact of these two countries’ extreme values.

The population confound. Given `data_centers` correlates with population at $r = 0.897$ (Section 3.3.3), the raw data centre effect’s robustness to population was tested directly by comparing its coefficient before and after population entered the model. Contrary to expectation, the coefficient did not weaken (e.g. 2025: $p = 0.001$ without population, $p = 0.003$ with population added); though this test is itself qualified by an asymmetry in how the two variables entered the model (`data_centers` continuous, population discretised into terciles for the reasons discussed in Section 3.3.3), which was confirmed directly by testing continuous (raw and log-transformed) population as an alternative, which both produced numerically unstable fits (inflated standard errors, non-significant population and data centre effects), consistent with the two variables’ known collinearity manifesting differently

depending on the resolution at which population enters the model.

Choice of country-size control. The complete final specification was refitted with land area in place of population as the size control (both discretised into terciles on identical grounds; Section 3.3.3). This is the single most consequential robustness check in this analysis. We saw the raw data centre count effect, which had survived every other check, loses statistical significance under every year when land area replaces population as the size control (e.g. 2025: $p = 0.001$ under population, $p = 0.185$ under land area). The price-gap and transitivity findings are unaffected by this substitution in every year. This result is treated as decisive: the raw data centre count finding is reported throughout this analysis as conditional on the choice of size control, not as an unconditionally robust result.

3.3.5 Cross-Sectional Regression

3.3.5.1 Motivation

The ERGM (Section 3.3.4) establishes that a country’s price relative to its trading partners, and its data centre count, are associated with tie *formation* in the trade network. This does not itself establish whether these same variables relate to the two country-level outcomes of substantive interest to this analysis, data centre presence and price, since the ERGM’s outcome is the network’s structure, not either of these variables directly. This subsection translates the ERGM’s dyadic findings into country-level predictors and tests their relationship with each outcome directly, using ordinary least squares (OLS) regression with countries as the unit of observation ($n = 29$).

3.3.5.2 Model Formulation

For an outcome y_i (price or data centre count) observed for country i , and a vector of predictors $\mathbf{x}_i$, the model is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \quad (3.8)$$

or, in matrix form, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{y}$ is the $n \times 1$ vector of outcomes, $\mathbf{X}$ the $n \times (p+1)$ design matrix (a column of ones for the intercept, followed by each predictor), $\boldsymbol{\beta}$ the $(p+1) \times 1$ vector of coefficients, and $\boldsymbol{\varepsilon}$ the $n \times 1$ vector of errors. Each β_k is interpreted as the expected change in y_i associated with a one-unit increase in x_{ki} , holding every other predictor in the model fixed.

Equation 3.8 makes explicit the assumptions OLS relies on: that errors are independent across observations, identically distributed, and homoskedastic (constant variance σ^2 , not varying with $\mathbf{x}_i$). The first of these (independence across observations) is precisely the assumption the ERGM in Section 3.3.4 was introduced to address for the *network* itself; it is a separate, and in this cross-sectional setting more defensible, assumption when applied to country-level outcomes directly, though it is revisited formally in Section 3.3.6, where the possibility that one country’s outcome depends on its trading partners’ realised outcomes (rather than only on its own covariates) is modelled explicitly.

3.3.5.3 Estimation Algorithm (OLS)

Unlike the ERGM, β in Equation 3.8 has a closed-form solution, obtained by minimising the residual sum of squares (RSS):

Algorithm: Ordinary Least Squares Estimation

1. **Assemble** the design matrix $\mathbf{X}$ and outcome vector $\mathbf{y}$.
2. **Minimise** $\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$, giving the closed-form solution:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (3.9)$$

3. **Estimate** $\hat{\sigma}^2 = \text{RSS}(\hat{\beta}) / (n - p - 1)$ and $\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1}$, from which standard errors follow.
4. **Test** each coefficient via $t_k = \hat{\beta}_k / \text{SE}(\hat{\beta}_k)$ against a t_{n-p-1} distribution.

Figure 3.7: OLS estimation via the normal equations (Hastie et al. 2009).

Unlike the ERGM, no iterative or simulation-based procedure is required. Equation 3.9 is an exact, direct solution given $\mathbf{X}$ and $\mathbf{y}$.

3.3.5.4 Network Exposure Variables

Because the ERGM’s findings concern properties of a *pair* of countries (a price gap, or the co-occurrence of two countries’ data centre levels), they must be translated into a property of a single country before they can enter Equation 3.8. For each country i , with $N(i)$ denoting its set of trading partners in the (static; Section 3.3.3) network:

$$\text{avg_price_gap_with_partners}_i = \frac{1}{|N(i)|} \sum_{j \in N(i)} |\text{price}_i - \text{price}_j| \quad (3.10)$$

$$\text{avg_price_of_partners}_i = \frac{1}{|N(i)|} \sum_{j \in N(i)} \text{price}_j \quad (3.11)$$

Equation 3.10 is the country-level analogue of the ERGM’s `absdiff(price)` term (Equation 3.5). A country whose trading partners have very different prices to its own will have a high value here. Equation 3.11 instead asks whether a country’s own price tracks the price environment of its neighbours, which is a standard spatial-lag exposure measure, formalised further in Section 3.3.6.

The two variables above are *not* interchangeable across outcomes; using Equation 3.10 to predict `price_eur_mwh` itself would be mechanically circular, since the gap is partly constructed from the country’s own price value. This is not a problem when the outcome is data centre count, a variable unrelated to the gap’s construction. For this reason, the data centre models use Equation 3.10 as a predictor, while the price model uses Equation 3.11 instead.

3.3.5.5 Specifications

$$\begin{aligned} \text{data_centers}_i \text{ (or } \text{dc_per_capita}_i) = & \beta_0 + \beta_1 \text{avg_price_gap_with_partners}_i \\ & + \beta_2 \log(\text{population}_i) + \beta_3 \text{in_eu}_i \\ & + \beta_4 \text{pct_renewable}_i + \beta_5^\top \text{incentives}_i + \varepsilon_i \end{aligned} \quad (3.12)$$

$$\begin{aligned} \text{price_eur_mwh}_i = & \beta_0 + \beta_1 \text{avg_price_of_partners}_i + \beta_2 \text{pct_renewable}_i \\ & + \beta_3 \text{pct_nuclear}_i + \beta_4 \text{demand_per_capita}_i \\ & + \beta_5 \log(\text{population}_i) + \beta_6 \text{in_eu}_i + \varepsilon_i \end{aligned} \quad (3.13)$$

Although `dc_per_capita` is already normalised by population, `log(population)` is retained as a predictor in both specifications rather than dropped from the per-capita model. Dividing the outcome by population removes the *mechanical* scale effect (larger countries generally having more of things), but does not guarantee that population is unrelated to the resulting *density* measure, as a country’s population could still be systematically associated with higher or lower data centre density per person for reasons unconnected to sheer size. It is acknowledged that this may seem statistically unintuitive, however due to the complexity of this context, there is value in testing this relationship, bearing this in mind.

`incentives` enters Equation 3.12 as a categorical factor (three dummy coefficients, collected in β_5), consistent with its treatment throughout this analysis as an ordinal, hand-coded index rather than a continuous scale. Population enters here in continuous log form, rather than the tercile discretisation required for the ERGM (Section 3.3.3), since OLS does not share the ERGM’s estimation-stability sensitivity to continuous, skewed covariates. Regulation variables and an interaction term were deliberately excluded from Equation 3.13 as the ERGM’s regulation signal concerned tie formation and data centre-specific incentive structures, not price-setting, and a leaner specification was preferred given the modest sample size ($n = 29$).

3.3.5.6 Key Results (2025)

Three findings recur across all three years (2023–2025; full year-by-year output in the project repository). First, `log(population)` is a strong, highly significant predictor of raw data-centre count in every year, but reverses sign and remains significant when the outcome is re-expressed per capita (e.g. $\hat{\beta} = 63.0$, $p < 0.001$ for the raw count in 2025, versus a significantly *negative* coefficient for per-capita density in the equivalent specification), indicating that smaller countries carry disproportionately higher data centre density relative to population. This sign reversal itself confirms the value of retaining `log(population)` in the per-capita specification, rather than assuming its relationship with the outcome disappears once the outcome is normalised. A natural concern, given `log(population)`’s dominant role, is whether the raw-count model’s reported R^2 is substantially attributable to population alone. A population-only model was tested and achieves $R^2 = 0.579$ (adjusted $R^2 = 0.563$), compared to $R^2 = 0.783$ (adjusted $R^2 = 0.724$) for the full specification. This is an incremental contribution of approximately 20 percentage points from the remaining predictors, confirmed as jointly significant

Term	Estimate	Std. Error	<i>p</i>
<i>Outcome: data centre count (raw)</i>			
Intercept	−941.31	174.57	< 0.001 ***
Avg. price gap with partners	−0.21	1.32	0.877
log(Population)	63.02	10.14	< 0.001 ***
EU membership	−50.12	33.30	0.147
% Renewable generation	0.54	0.57	0.349
Incentives: level 2	67.12	26.21	0.018 *
Incentives: level 3	155.04	45.83	0.003 **
<i>Outcome: wholesale price (EUR/MWh)</i>			
Intercept	101.92	34.79	0.008 **
Avg. price of partners	0.67	0.12	< 0.001 ***
% Renewable generation	−0.30	0.12	0.023 *
% Nuclear generation	−0.47	0.12	0.001 **
Demand per capita	−2.03	0.55	0.001 **
log(Population)	−1.65	1.57	0.302
EU membership	−7.11	4.92	0.162

Table 3.4: Cross-sectional OLS results, 2025 ($n = 29$). data centre model: $R^2 = 0.783$, adjusted $R^2 = 0.724$, $F(6, 22) = 13.21$, $p < 0.001$. Price model: $R^2 = 0.875$, adjusted $R^2 = 0.841$, $F(6, 22) = 25.65$, $p < 0.001$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

via a nested F -test ($F(5, 22) = 4.13$, $p = 0.008$). Population is unambiguously the single most influential predictor, but the raw-count model’s overall fit is not attributable to population alone, as the remaining covariates, taken together, add a statistically significant amount of explanatory power beyond country size.

Second, the highest incentive-regulation category is a significant, consistently positive predictor of data centre count (raw: $\hat{\beta}_{\text{level 3}} = 82.5, 124.6, 155.0$ across 2023–2025, all $p < 0.02$), though this category contains only two countries and is examined in detail in the robustness section below. Third, a country’s price level tracks the average price of its trading partners closely and significantly in every year ($\hat{\beta} = 0.58\text{--}0.86$, $p < 0.001$), a spatial convergence pattern formally tested in Section 3.3.6. The price-gap exposure variable (Equation 3.10) is a clean null result which also recurs throughout and shows no relationship with either data centre outcome in any year ($p = 0.44\text{--}0.88$ throughout); the ERGM’s tie-formation finding for price gap does not extend to explaining data centre presence.

3.3.5.7 Robustness Checks

Collinearity and influence diagnostics. VIFs (see 3.3.3), were below 2 for every predictor in every model, within conventional thresholds (O’Brien 2007). This is interpreted as reassurance against the most severe forms of multicollinearity, but not as proof of its absence, despite our extensive covariate screening (Section 3.3.3). Cook’s distance,

$$D_i = \frac{e_i^2}{p \cdot \text{MSE}} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}, \quad (3.14)$$

(where e_i is the i -th residual, h_{ii} the corresponding leverage, and p the number of parameters) was computed for every model against the conventional $4/n$ threshold. This flagged the

countries occupying the thin **incentives** categories (Germany, United Kingdom, Ireland) consistently in the data centre models, motivating the two exclusion analyses below. The price model’s flagged countries varied across years (Macedonia, Switzerland, Sweden, Norway, Italy, Serbia, Greece), with no single country recurring, indicating this model is not dominated by any one observation.

Decomposing the incentives effect. The highest incentive-regulation category contains exactly two countries, the United Kingdom and Ireland. Excluding each in turn from the 2025 per-capita specification shows the effect is driven almost entirely by Ireland. Excluding the United Kingdom leaves the coefficient intact and strengthened ($\hat{\beta} = 16.65$, $p < 0.001$), while excluding Ireland eliminates it entirely ($\hat{\beta} = 3.26$, $p = 0.417$). Ireland’s per-capita data centre density more than quadrupled between 2023 and 2025 (4.71 to 22.06), while the United Kingdom’s grew only modestly (4.06 to 6.02) over the same period. This finding is interpreted as an illustrative single-country case, consistent with Ireland’s well-documented status as a favourable jurisdiction for data centre investment, rather than a generalisable effect of high-incentive regulation as a category.

Sensitivity to the most influential country. Since $\log(\text{population})$ is the most consequential finding in this section, the raw-count specification was refitted excluding Germany (the country with the highest Cook’s distance in every year). The coefficient shrinks modestly (e.g. 63.0 $\rightarrow$ 49.9 in 2025) but remains highly significant in every year ($p < 0.001$), confirming the finding is not an artefact of Germany’s influence.

3.3.6 Network Autocorrelation Model (Spatial Autoregressive Model)

3.3.6.1 Motivation

The cross-sectional regressions in Section 3.3.5 test network dependence using an engineered exposure variable (Equation 3.11), treating the average value of a country’s trading partners as an ordinary, independent regressor. This approach has a specific limitation in that it ignores *reflexivity*. A country’s price plausibly depends on its trading partners’ prices, but those partners’ prices are simultaneously, and in part, determined by trading with that same country. Treating the partner-average as an exogenous predictor, as the OLS specification in Section 3.3.5 does, does not account for this feedback loop, and produces a biased estimate of how strongly network position actually predicts the outcome.

A Network Autocorrelation Model (Doreian 1980) (most commonly estimated as a Spatial Autoregressive (SAR) model, the term used throughout this section) addresses this directly by estimating the strength of network dependence *jointly* with the rest of the model’s coefficients via maximum likelihood, rather than treating a pre-computed neighbour average as an ordinary covariate. This is the same underlying idea as the exposure variable used in Section 3.3.5, but formalised and corrected for the reflexivity problem described above. The two are directly comparable, and the SAR model is used here specifically as the more rigorous test of the same substantive question the exposure variable was built to answer.

3.3.6.2 Model Formulation

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (3.15)$$

where $\mathbf{y}$ is the $n \times 1$ vector of outcomes (price, or data centre count, one value per country); $\mathbf{W}$ is the $n \times n$ *spatial weights matrix*, here the trade network's adjacency matrix, row-standardised so that each row sums to 1 (so that $\mathbf{W}\mathbf{y}$ is, for each country, the average outcome among its trading partners); $\mathbf{X}$ and $\boldsymbol{\beta}$ are the design matrix and coefficient vector exactly as in Equation 3.8; and $\boldsymbol{\varepsilon}$ is the error term. The parameter of central interest is ρ (the *spatial autoregressive coefficient*), which measures the strength of the outcome's dependence on the outcome of a country's own trading partners, net of every covariate in $\mathbf{X}$: $\rho = 0$ recovers the standard OLS model of Equation 3.8 exactly, with no network dependence in the outcome; $\rho > 0$ indicates that higher partner outcomes are associated with higher own outcomes, beyond what $\mathbf{X}$ alone predicts.

Equation 3.15 makes the reflexivity problem explicit, with $\mathbf{y}$ appearing on both sides, since $\mathbf{W}\mathbf{y}$ depends on the very outcomes being modelled for every country's trading partners. Solving Equation 3.15 for $\mathbf{y}$ gives the model's *reduced form*:

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \quad (3.16)$$

which shows that a shock to any one country's error term ε_i propagates, through $(\mathbf{I} - \rho \mathbf{W})^{-1}$, to every other country connected to it through the network; precisely the feedback structure that a plain OLS regression using $\mathbf{W}\mathbf{y}$ as an ordinary covariate cannot capture, since OLS treats $\mathbf{W}\mathbf{y}$ as fixed and exogenous rather than as a function of the same error term appearing on the left-hand side.

3.3.6.3 Estimation Algorithm (Maximum Likelihood)

Because of the reflexivity in Equation 3.15, ρ and $\boldsymbol{\beta}$ cannot be estimated by ordinary least squares. Instead, both are estimated jointly by maximum likelihood, exploiting the reduced form in Equation 3.16.

Algorithm 3.8 is implemented directly via `spatialreg::lagsarlm()`:

```
listw_2025 <- spdep::mat2listw(adj_matrix, style = "W")
sar_price_2025 <- spatialreg::lagsarlm(
  price_eur_mwh ~ pct_renewable + pct_nuclear + demand_per_capita +
    log_population + in_eu,
  data = country_data_2025,
  listw = listw_2025)
```

where `adj_matrix` is the network's row-ordered adjacency matrix (Section 3.3.3). The likelihood ratio test (Equation 3.18) is computed via `lmtest::lrtest()`, comparing this fit against the equivalent `lm()` specification with ρ fixed at 0.

Algorithm: Maximum Likelihood Estimation for the SAR Model

1. **Construct** the row-standardised weights matrix $\mathbf{W}$; confirm no isolated nodes.
2. **Write** the log-likelihood, including the Jacobian term $\ln |\mathbf{I} - \rho \mathbf{W}|$:

$$\begin{aligned} \ell(\rho, \boldsymbol{\beta}, \sigma^2) = & -\frac{n}{2} \ln(2\pi\sigma^2) + \ln |\mathbf{I} - \rho \mathbf{W}| \\ & - \frac{1}{2\sigma^2} [(\mathbf{I} - \rho \mathbf{W})\mathbf{y} - \mathbf{X}\boldsymbol{\beta}]^\top [(\mathbf{I} - \rho \mathbf{W})\mathbf{y} - \mathbf{X}\boldsymbol{\beta}] \end{aligned} \quad (3.17)$$

3. **Concentrate** the likelihood: for any ρ , $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ have closed-form OLS-like solutions (Equation 3.9 applied to $(\mathbf{I} - \rho \mathbf{W})\mathbf{y}$), reducing to a function of ρ alone.
4. **Maximise** the concentrated log-likelihood over the single parameter ρ via one-dimensional numerical search.
5. **Recover** $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}^2$ at the maximising $\hat{\rho}$; obtain standard errors from the inverse Hessian.

Figure 3.8: Maximum likelihood estimation of the SAR model, following the concentrated-likelihood approach of (Ord 1975), as implemented in `spatialreg::lagsarlm()` (formerly part of `spdep`) (Bivand & Piras 2015), an estimator validated against alternative software implementations in the cited comparison.

3.3.6.4 Testing Whether ρ s Needed: the Likelihood Ratio Test

Before interpreting $\hat{\rho}$, whether the network-lag term improves the model at all is tested formally via a likelihood ratio (LR) test against the corresponding non-spatial OLS model (i.e. Equation 3.15 with ρ constrained to 0):

$$\text{LR} = 2(\ell_{\text{SAR}} - \ell_{\text{OLS}}) \sim \chi_1^2 \quad (3.18)$$

under the null hypothesis $\rho = 0$, since the SAR model adds exactly one parameter relative to its OLS-equivalent null.

Outcome	Year	LR statistic	p
Price	2023	14.88	< 0.001 ***
Price	2024	12.93	< 0.001 ***
Price	2025	21.64	< 0.001 ***
Data centres (raw)	2023	1.22	0.270
Data centres (raw)	2024	0.45	0.501
Data centres (raw)	2025	0.48	0.488
Data centres (per capita)	2023	0.01	0.922
Data centres (per capita)	2024	0.07	0.789
Data centres (per capita)	2025	0.01	0.940

Table 3.5: Likelihood ratio tests (Equation 3.18) comparing each SAR model against its non-spatial OLS-equivalent null, all three years. *** $p < 0.001$.

The results in Table 3.5 split cleanly by outcome. For price, the network-lag term is highly significant in every year, indicating that the network’s structure explains variation in price beyond what generation mix, demand, population, and EU membership account for

alone. For data centre count, in both raw and per-capita form, the network-lag term is clearly non-significant in every year. This formally confirms, using a direct test of the outcome’s own network dependence rather than the price-based proxy used in Section 3.3.5, that data centre location shows no detectable network dependence, while price does.

The likelihood ratio test (Table 3.5) is the primary result reported for this stage of the pipeline outlined in Figure 3.4, directly answering the question motivating the SAR model’s use (does network position explain variation in the outcome beyond the covariates already in $\mathbf{X}$), and does so without requiring $\hat{\rho}$ ’s point estimate to be interpreted on its own. Full coefficient output for every SAR model is available in the project repository.

3.3.6.5 Robustness and Limitations

As with every model in this analysis, $n = 29$ is modest, and ρ ’s standard error relies on asymptotic theory (the inverse Hessian, Step 5 in Algorithm 3.8) that is more reliable in larger samples, so significance tests on ρ are interpreted with this in mind. A further, distinct limitation is documented negative bias in MLEs of ρ specifically in network autocorrelation models (Wang et al. 2014), whereby any significant $\hat{\rho}$ reported here is more likely to understate the true strength of network dependence than to overstate it. This means that a significant result found despite this bias reinforces the finding that price shows genuine network dependence. Although, the magnitude of $\hat{\rho}$ itself should not be over-interpreted. This limitation does not undermine the central conclusion of this subsection, however, since the result is not a single borderline p -value but a consistent pattern across three independent years. The price models’ LR statistics are an order of magnitude larger than any data centre model’s, and every price-model LR test clears $p < 0.001$, while every data centre model sits at $p > 0.27$. A finding this large and this consistently one-sided is unlikely to be an artefact of small-sample unreliability in ρ ’s standard error specifically, since that concern would be expected to inflate uncertainty in both sets of models symmetrically, not to produce this scale of separation between them. The result should nonetheless be read alongside the sample-size caveat stated throughout this analysis, rather than as fully immune to it.

3.3.7 Summary of Findings from Network Analysis

Network structure has a strong, formally confirmed role in explaining price. A country’s wholesale price is significantly predicted by its trading partners’ prices, beyond what generation mix, demand, population, and EU membership explain alone (LR test $p < 0.001$ in every year). This confirms network structure as a statistically significant explanatory factor for price. Consequently, any model using price to predict data centre presence without also controlling for network structure risks omitted-variable bias, since price is not independent of a country’s trade-network position.

Network structure has no detectable role in explaining data centre presence, in either raw or per-capita form. Neither the price-gap exposure variable nor a formal spatial-lag term linking a country’s own data centre level to its neighbors’ shows any relationship. data centre location instead appears driven by country-level structural factors

(population size, incentive regulation, most specifically in Ireland’s case) rather than network position.

In sum of this analysis, **European energy flow network structure matters for energy price, not as much for where data centres locate**. The two outcomes we set out to explain have different relationships with the trade network. This nuanced result is a product of a careful, multi-method investigation (ERGGM + OLS + SAR) which involved robustness checks at every stage, either for correlated variables or for exclusion-sensitivity for extreme/driving cases (important given our small sample size of $n=29$ countries).

Incentives regulation was the one regulation variable to show a significant, consistent relationship with data centre presence across all three years, in both the raw-count and per-capita cross-sectional models (raw: $\hat{\beta}_{\text{incentives}=3} = 82.5, 124.6, 155.0$ for 2023–2025, all $p < 0.02$). However, this category contains only two countries, the United Kingdom and Ireland, and closer examination reveals the effect is driven almost entirely by Ireland. Ireland’s per-capita data centre density more than quadrupled between 2023 and 2025 (4.71 to 22.06), while the United Kingdom’s grew only modestly (4.06 to 6.02). When the two countries are tested separately, excluding the United Kingdom leaves the effect intact and strengthened ($\hat{\beta} = 16.65$, $p < 0.001$), while excluding Ireland eliminates it entirely ($\hat{\beta} = 3.26$, $p = 0.42$). This is consistent with Ireland’s well-documented status as a favourable jurisdiction for data centre investment, attributed in the industry literature to its corporate tax regime and supportive regulatory environment. We therefore interpret this with caution as an illustrative single-country case rather than a generalisable effect of high-incentive regulation as a category.

Population shows a striking reversal depending on how data centre presence is measured. Larger countries have significantly more data centres in absolute terms, but significantly fewer per capita, whilst smaller countries carry a disproportionately high data centre density relative to their population. This finding was robust to excluding the most influential country in the sample (Germany) in both the ERGGM and the cross-sectional regressions.

3.3.8 Key Limitation

The analysis relies on a network structure that is fixed across the study period and on regulatory classifications that likewise do not change year to year, so any observed association (most concretely the Ireland case, where high-incentive regulation and rapidly rising data centre density coincide) is equally consistent with the structural factor causing the outcome, the outcome having prompted the structural factor (e.g. incentives introduced in response to an already-growing sector), or both being driven by a common unobserved cause. The small sample size ($n=29$ countries) and thin regulatory categories compound this limitation further, since even the associational estimates themselves rest on limited variation and, in some cases, a handful of influential observations. Therefore, none of these relationships should be interpreted causally. The next section employs different techniques to inform causal conclusions regarding what factors drive data centre presence.

3.4 Causal Analysis

3.4.1 Background

Causal analysis represents the natural progression of statistics, moving past the paradigm of identifying coincidences and instead challenging the notion that data can indeed represent and aid in computing counterfactuals. The field of causal analysis and causal inference has seen major developments in the 21st century, and finds roots in qualitative frameworks, built via human intuition, and uses data to reinforce or challenge our human beliefs.

Pearl & Mackenzie (2018) highlight in ‘The Book of Why’ the key developments required to bridge the gap between a classical approach to data modelling, in the way of regression, to a causal approach to understanding data. In the book, Pearl coins the ‘Ladder of Causation’ consisting of three rungs representing distinct levels of cognitive ability; seeing, doing, and imagining. In his writings, Pearl also explains many causal relationships through Directed Acyclic Graphs (DAGs), which is a visual aid that can represent technical data, while maintaining a level of explainability easily digestible by a non-technical audience. The usage of these graphs permeates throughout the literature surrounding causal analysis and will likewise be consistently used in this paper.

This section aims to construct and execute a coherent causal analysis pipeline to identify and estimate the average treatment effect (ATE) and conditional average treatment effect (CATE) of various treatment variables on the intended outcome, that being **Data Centers**. The pipeline commences with minor data manipulation, as an extension of the previous methodology, to function in the context of causal graphs, and proceeds with the application of the DoWhy Framework, a Python library developed by Sharma & Kiciman (2020) on behalf of the Microsoft Research team, and eventually extended to the DoWhy-GCM for Graphical Causal Models (GCM), authored by Blöbaum et al. (2024) with Amazon Web Services. The DoWhy Framework introduces a four-step pipeline to model, identify, estimate, and refute causal effects of treatments, the details of which will be covered in its respective section.

Following the initial pipeline for causal understanding, the study will return to a more traditional approach, employing regression techniques to allow for estimation with the causal understanding in mind. Regression analysis applied in a causal context allows for the causal relationship between variables to come under consideration when determining what would traditionally be understood as ‘correlation’ between variables. The SAGE Handbook of Regression Analysis and Causal Inference authored by Best & Wolf (2014) provides an excellent foundation for two such regression techniques: Panel Regression with a Fixed-Effects Model, and Instrumental Variables (IV) Regression. Covered in detail in Chapters 13 and 15, the handbook provides the reader with the knowledge and requisite assumptions called upon by each model, accompanied by the aforementioned causal DAGs to explain the relationships between treatment and outcome variables. The paper will also discuss strengths and limitations of the respective models and why they may or may not be appropriate for this context.

In alignment with the objective of this paper, techniques and models used within causal analysis and causal inference will be borrowed and applied in the context of energy usage and data centre growth in Europe. The methods introduced in this section, and the mechanisms

by which they function, will be elaborated upon as they come up.

3.4.2 Introducing the DoWhy Framework

Understanding the links made through initial data explorations is the natural next step in this causal analysis pipeline. This process, dubbed ‘causal understanding’, has seen major developments in recent times, and this paper will make use of a Python library developed by the Microsoft Research team called the DoWhy library to programmatically aid in the causal inference process, and confirm or refute our beliefs embedded in the causal model constructed through the regression analysis. The paper ‘DoWhy: An End-to-End Library for Causal Inference’ is a product of the Microsoft Research team spearheaded by authors Sharma & Kiciman (2020), which provides an easy interface for analysts to follow the best practices of causal inference, which builds around four key steps: Model, Identify, Estimate, and Refute.

Further supporting the usage of this library, in 2024 researchers Blöbaum et al. (2024) with Amazon Web Services contributed to an extension of the library, called DoWhy-GCM. In their paper ‘DoWhy-GCM: An Extension of DoWhy for Causal Inference in Graphical Causal Models’, the authors extend the framework to a graphical approach, which helps to answer a ‘broader range of causal questions’. The library enables users to ask more causal questions about suggested DAG models and identify root causes of changes in the data.

Evidence of the library’s application in the field of energy modelling is apparent. Authors Jiang et al. (2026) in the paper ‘Causal inference and machine learning in power and energy systems: A review of assumptions, applications and opportunities’, and Chauhan et al. (2026) in the paper ‘Causal modelling of energy and resource demand’, have showcased the capabilities of the DoWhy library in identifying causal links within and surrounding energy systems. Jiang focuses largely on individual climate knowledge, both measured and perceived, and how it influences the carbon footprint of an individual. The study indicates that simply ‘increasing factual climate knowledge can meaningfully contribute to lowering carbon footprints’, establishing a clear causal relationship between the treatment and the outcome, while controlling for confounding factors. Chauhan approaches the issue of energy modelling through three overarching lenses, transport, electricity, and water. Chauhan showcases the graphical capabilities of the DoWhy library, and uses the tools provided in the DoWhy-GCM extension to learn and validate a causal DAG between several treatments, outcomes, and confounders.

Accompanying the introduction of the DoWhy library is a research paper outlining the four key steps of an end-to-end causal analysis pipeline. These four steps will be applied to this study in the context of data centre growth. Authors Sharma & Kiciman (2020) of the DoWhy research paper included a graphic to illustrate the process of identifying causal effects based on data and domain knowledge, shown below:

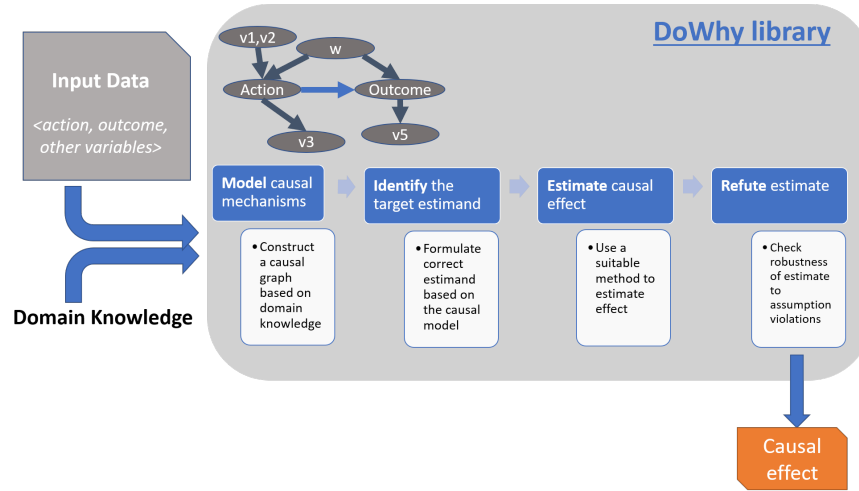


Figure 3.9: The four-step analysis pipeline in DoWhy

The four steps outlined in the paper, model, identify, estimate, and refute, are essential to complement the causal intuition surrounding data centres based on prior domain knowledge and supporting literature.

3.4.3 Data Preparation

Working with the dataset described in 3.3.3, the merging of several electricity generation types was necessary for ease in causal analysis. Electricity generation by fuel type exhibits high collinearity, since a lack of generation in one category means high generation in another. Controlling for the number of these variables contained in the model is key to avoiding spurious relationships. The data sources provided by Ember (2026) outlines these fuel generation group, which will be aggregated to indicate electricity generation by Fossil Fuels, Nuclear, and Renewables. The chart below indicates which fuel generation type falls into its respective category:

Demand									
Total Generation									
Clean									
Renewables					Nuclear	Fossil			
Wind and Solar		Hydro, Bioenergy and Other Renewables				Gas and Other Fossil			
Wind	Solar ¹	Hydro ²	Bioenergy ³	Other Renewables ⁴		Coal	Gas	Other Fossil ⁵	

¹ Solar includes both solar thermal and solar photovoltaic generation, and where possible distributed solar generation is included.

² Where possible, Hydro generation excludes any contribution from pumped hydro generation.

³ Bioenergy is classified as renewable, but caveats are attached. See below for details.

⁴ Other Renewables generation includes geothermal, tidal and wave generation.

⁵ Other Fossil generation includes generation from oil and petroleum products, as well as manufactured gases and waste.

Figure 3.10: Breakdown of Fuel Types in their respective categories in the Ember Dataset

Then comes the concern of treatment variables. As the aim of this causal analysis study is

to provide insight to policy, which implies actionable steps to change the variables being outlined. For this reason, certain variables, such as the geographic variables of **Population** and **Land mass**, will be placed under more scrutiny, since the correlation and causation between these variables and the outcome of **Data Centers** are trivial and not easily influenced through changes in policy. An extension discussing these geographic variables is included towards the end of this section. Thus the main treatments being tested are the electricity generation types, **Renewables_Generation**, **Fossil_Generation**, and **Nuclear_Generation**, as well as the regulatory frameworks, **reporting**, **sustainability**, **incentives**, and **planning**.

3.4.4 Causal Inference Pipeline

3.4.4.1 Modelling the Causal Question

The DoWhy analysis pipeline suggests a causal graph be constructed based on prior domain knowledge, to establish what researchers believe to be the prevailing cause and effect in the real world. The variables included in this study is an abstracted variant of those laid out in the above papers, and several proxies have been used in modelling certain variables when direct data could not be located.

The DoWhy library also supports the construction of such causal graphs with built-in methods for causal discovery to learn these causal graphs. Three such methods are showcased in the DoWhy documentation, those being the Peter-Clark (PC) algorithm, Greedy Equivalence Search (GES), and the Linear Non-Gaussian Acyclic Model (LiNGAM) (Shimizu et al. 2006). Unlike PC and GES, which typically recover a completed partially directed acyclic graph (CPDAG), LiNGAM aims to recover a fully directed graph under the stronger assumption that the data are generated by a linear structural equation model with mutually independent, non-Gaussian disturbances and no latent confounders. This makes LiNGAM particularly attractive for constructing a preliminary DAG that can subsequently be refined using domain knowledge. For this reason, the LiNGAM method is preferred to learn a preliminary structure for the causal DAG, and will be manually adjusted based on qualitative understanding and intuition of the variables involved. The preliminary graph provided by the LiNGAM method is shown below:

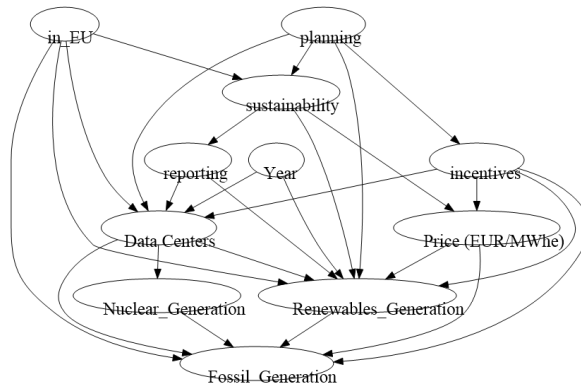


Figure 3.11: Preliminary DAG from the Linear Non-Gaussian Acyclic Model (LiNGAM)

The LiNGAM Mechanism. The LiNGAM algorithm was applied to the panel dataset described earlier, treating each country-year observation as a sample. The resulting graph provides an initial hypothesis for causal directions between variables, then refined by enforcing domain constraints such as the exogeneity of geographical variables (e.g. land area) and the temporal ordering of demand-side variables (e.g. population and data centres). The LiNGAM method has seen many developments in the past years, and several variants exist to learn the causal DAG. Supported by the DoWhy library are the Independent Component Analysis (ICA) based method, DirectLiNGAM (Shimizu et al. 2011), Repetitive Causal Discovery (RCD) (Maeda & Shimizu 2020), and Vector Autoregression (VAR) (Hyvärinen et al. 2010).

In this study, the ICA-based variant of LiNGAM is used. Under the model $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, ICA estimates an unmixing matrix W such that $\mathbf{e} = W\mathbf{x}$. Because ICA identifies the sources only up to permutation and scaling, LiNGAM resolves these ambiguities by selecting the unique row permutation that produces a non-zero diagonal and then normalising the rows so that the diagonal entries are unity. The causal matrix is then recovered as $\hat{B} = I - W'$, and the resulting structure is interpreted as an initial candidate DAG. This graph is then manually adjusted to respect subject-matter constraints such as the exogeneity of land area and the expected temporal ordering of policy, energy, and demand variables.

LiNGAM model. Let p denote the number of observed variables and let

$$\mathbf{x} = (x_1, \dots, x_p)^\top \in \mathbb{R}^p \quad (3.19)$$

be the observed data vector for a single country-year observation. The LiNGAM model assumes that each variable x_i is generated by a linear structural equation of the form

$$x_i = \sum_{j=1}^p b_{ij}x_j + e_i + \mu_i, \quad i = 1, \dots, p, \quad (3.20)$$

where b_{ij} is the causal effect of x_j on x_i , e_i is an exogenous disturbance term, μ_i is a constant intercept, and $b_{ij} = 0$ whenever x_j is not a direct cause of x_i . Stacking the equations gives

$$\mathbf{x} = B\mathbf{x} + \mathbf{e} + \boldsymbol{\mu}, \quad (3.21)$$

where $B = (b_{ij})$, $\mathbf{e} = (e_1, \dots, e_p)^\top$, and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_p)^\top$. After centering the data, the intercept term may be ignored, yielding

$$\mathbf{x} = B\mathbf{x} + \mathbf{e}. \quad (3.22)$$

Rearranging,

$$(I - B)\mathbf{x} = \mathbf{e}, \quad \mathbf{x} = (I - B)^{-1}\mathbf{e}. \quad (3.23)$$

Define

$$A = (I - B)^{-1}, \quad (3.24)$$

so that the model becomes

$$\mathbf{x} = \mathbf{A}\mathbf{e}. \quad (3.25)$$

Thus, the observed variables are a linear mixture of the independent disturbances.

Alternative LiNGAM variants were considered. DirectLiNGAM estimates the causal ordering directly from residual independence without using ICA, while VARLiNGAM extends the framework to time series by separating lagged and contemporaneous effects. However, because the present data contain only three annual observations per country, the time dimension is too short to make VARLiNGAM a robust primary discovery strategy. For this reason, ICA-LiNGAM provides the best balance between interpretability, identifiability, and compatibility with the available data.

This provides a good baseline understanding for how variables in the dataset might interact with one another, and with the intended outcome variable; **Data Centers**. However, there are certain links that should be manually overridden to ensure that it aligns with the qualitative understanding provided by the literature, and by intuition. For example, the causal link between **Data Centers** and **Price (EUR/MWhe)** was added to align with the literature. This process must be done carefully to also ensure that the acyclic nature of the graph is preserved. This results after this adjustment is shown in the final candidate model DAG below:

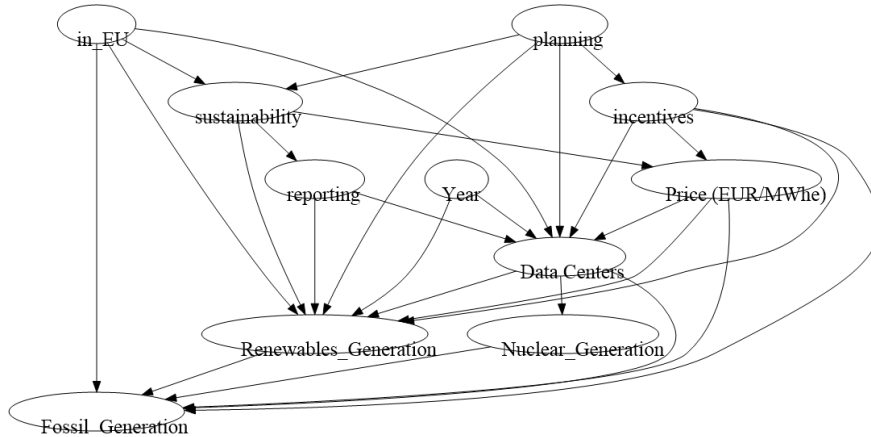


Figure 3.12: Final Adjusted DAG from the Linear Non-Gaussian Acyclic Model (LiNGAM)

From this graph, we can identify the variables and split them based on their relationship with the outcome variable: **Data Centres**. Within causal analysis, variables receive different categorisations based on their interaction with the outcome, with the most relevant categories being the confounders, mediators, and colliders.

Refuting the Causal Graph. Before proceeding with subsequent analyses, the validity of the causal graph should be tested to ensure that the graph is consistent with the data, verifying that the LiNGAM method provided a satisfactory graph, and the alterations made are still coherent with the initial data.

The falsification procedure in DoWhy-GCM evaluates whether the conditional independencies implied by the DAG are compatible with the observed data. Concretely, it tests, for each node, whether the node is independent of its non-descendants conditional on its parents at a significance level of 0.05. In our case, none of these local Markov conditions were rejected, suggesting that the adjusted LiNGAM DAG provides a statistically plausible representation of the dependence structure in the data.

The DoWhy library streamlines the process of graph refutation by checking Local Markov Conditions (LMCs), and performing independence tests (at a 0.05 significance level) between all LMCs, conditioning on all implied constraints in the graph. Performing this independence test, the following results are obtained:

```
+-----+
|                                     |
|                               Falsification Summary                       |
|-----+-----+
| The given DAG is informative because 0 / 20 of the permutations lie in the Markov |
| equivalence class of the given DAG (p-value: 0.00).                         |
| The given DAG violates 10/40 LMCs and is better than 100.0% of the permuted DAGs (p-value: 0.00). |
| Based on the provided significance level (0.05) and because the DAG is informative, |
| we do not reject the DAG.                                                  |
|-----+-----+
```

Figure 3.13: Independence Testing of LMCs in the DAG

This result verifies that the independence tests conducted on the LMCs satisfies the level of significance and is not rejected. Thus, the analysis can proceed with the DAG.

3.4.4.2 Identifying the Causal Estimand

The causal effects of the candidate treatments are estimated using the DoWhy framework, which separates the causal analysis into four steps: model, identify, estimate, and refute. In this section, the DAG learned from the LiNGAM procedure is supplied as the structural model, and DoWhy is then used to identify the estimand implied by the graph, estimate the corresponding causal effect, and test the robustness of the estimate under placebo interventions.

Average Treatment Effect. Let T denote a treatment variable, Y the outcome variable **Data Centers**, and Z the adjustment set implied by the DAG. Under the backdoor criterion, Z blocks all non-causal paths between T and Y , so that the causal effect of T on Y is identifiable from the conditional outcome distribution $P(Y | T, Z)$. In the simplest case, the average treatment effect (ATE) may be written as

$$\text{ATE}(T) = \mathbb{E}[Y(1) - Y(0)], \quad (3.26)$$

where $Y(1)$ and $Y(0)$ denote the potential outcomes under treatment and control respectively. More generally, for a treatment with multiple levels or a continuous treatment, the ATE may be interpreted as the expected change in Y induced by a unit change in T , averaged over the target population.

The conditional average treatment effect (CATE) extends this idea by conditioning on

covariates X , and is defined as

$$\text{CATE}(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]. \quad (3.27)$$

Whereas ATE summarizes the mean effect across the whole population, CATE captures treatment-effect heterogeneity across subgroups defined by observed features. To estimate the causal effect, DoWhy fits a backdoor-adjusted regression model of the form

$$Y_i = \beta_0 + \beta_1 T_i + \boldsymbol{\beta}^\top Z_i + \varepsilon_i, \quad (3.28)$$

where the coefficient β_1 is interpreted as the causal effect of the treatment after conditioning on the adjustment set Z . In the present implementation, the backdoor estimator is computed using linear regression, and statistical significance is assessed using the built-in significance test provided by DoWhy.

3.4.4.3 Estimating Causal Effect

Backdoor Estimation Criterion. Under the backdoor criterion, the causal effect of T on Y is identified by adjusting for a set of observed covariates Z such that every backdoor path from T to Y is blocked. This yields the backdoor adjustment formula

$$\mathbb{E}[Y \mid \text{do}(T = t)] = \int \mathbb{E}[Y \mid T = t, Z = z] dP(Z = z), \quad (3.29)$$

or, in the discrete case,

$$\mathbb{E}[Y \mid \text{do}(T = t)] = \sum_z \mathbb{E}[Y \mid T = t, Z = z] P(Z = z). \quad (3.30)$$

The average treatment effect between two treatment levels t_1 and t_0 is then

$$\text{ATE}(t_1, t_0) = \mathbb{E}[Y \mid \text{do}(T = t_1)] - \mathbb{E}[Y \mid \text{do}(T = t_0)]. \quad (3.31)$$

DoWhy operationalises this by fitting a regression of the form

$$Y = f(T, Z) + \varepsilon, \quad (3.32)$$

where f is chosen here to be linear.

3.4.4.4 Refuting the Causal Effects

Placebo refutation. After estimating each treatment effect, the robustness of the estimate is assessed using the placebo treatment refuter. This method replaces the original treatment variable T with an independent random placebo variable T^* , then re-estimates the causal effect. Under a valid causal specification, the placebo variable should have no systematic effect on the outcome, so the re-estimated effect should be close to zero. Formally, if the

original estimator is valid, then

$$\mathbb{E}[Y \mid do(T^*)] \approx \mathbb{E}[Y], \quad (3.33)$$

and the estimated placebo effect should not differ materially from zero. A large p-value in the placebo refutation test indicates that the estimator is not producing spurious effects under randomised treatment assignment, which provides support for the original causal specification.

3.4.4.5 Results for Estimating Causal Effects

Running the DoWhy pipeline on the set of treatments allows for the computation of the Average Treatment Effect (ATE) under the backdoor criterion, the adjustment set of observed covariates outlined in the backdoor estimation criterion, and the validity of the causal estimate, post-refutation via the placebo method. The specific details of each result is shown in the table below:

Treatment	Adjustment set	ATE	p-value	Placebo refutation
Price (EUR/MWhe)	incentives, sustainability	-0.70	0.121	Passed ($p = 0.94$)
in_EU	incentives, planning	-25.83	0.152	Passed ($p = 0.88$)
reporting	sustainability	41.91	0.186	Passed ($p = 0.86$)
sustainability	planning, in_EU	30.07	0.040	Passed ($p = 0.84$)
planning	in_EU	82.71	0.037	Passed ($p = 0.96$)
incentives	planning	56.97	0.006	Passed ($p = 1.00$)
Renewables_Generation	–	0.00	–	No directed path
Nuclear_Generation	–	0.00	–	No directed path
Fossil_Generation	–	0.00	–	No directed path

Table 3.6: DoWhy backdoor ATE estimates and placebo refutation results

The DoWhy backdoor estimators produced several treatment-specific average treatment effects. Among the policy variables, **incentives** and **planning** exhibited the strongest positive and statistically significant effects on **Data Centers**, with estimated ATEs of 56.97 ($p = 0.006$) and 82.71 ($p = 0.037$), respectively. **sustainability** was also positive and statistically significant, with an estimated ATE of 30.07 ($p = 0.040$). By contrast, **Price (EUR/MWhe)** and **in_EU** showed negative but statistically insignificant effects, while **reporting** had a positive but imprecisely estimated effect.

The placebo refuter supported the validity of the estimators used in these analyses. For all estimated treatments, the placebo effect was close to zero and the p-values were large (typically $p > 0.8$), indicating that the estimation procedure does not generate strong effects under randomised treatments. This provides some reassurance that the reported treatment effects are not simply artifacts of the regression specification.

3.4.5 Applying Causal Understanding to Regression Models

To complement the causal estimands identified using DoWhy, a set of regression models is estimated on the 2025 cross-section and on the full panel. Unlike the causal estimators, these regressions are not intended to recover the interventional effect directly; rather, they provide a compact linear summary of the observed association structure implied by the causal graph.

For the cross-sectional specification, let Y_i denote the number of data centres in country i , and let X_i be a vector of causally related predictors selected from the DAG. The OLS model is

$$Y_i = \beta_0 + \beta^\top X_i + \varepsilon_i, \quad (3.34)$$

where β is estimated by ordinary least squares, and the coefficient on each regressor measures its partial linear association with the outcome, conditional on the other included variables.

For the panel specification, let Y_{it} denote the outcome for country i in year t , and let W_{it} be a vector of time-varying predictors. The fixed-effects panel model is

$$Y_{it} = \alpha_i + \gamma_t + \theta^\top W_{it} + u_{it}, \quad (3.35)$$

where α_i captures time-invariant country-specific heterogeneity and γ_t captures common shocks affecting all countries in a given year. The parameter vector θ is identified only from within-country variation over time.

It is important to emphasize that these coefficients should not be interpreted mechanically as causal effects unless the adjustment set is valid and the causal assumptions encoded in the DAG are satisfied. In particular, a small or insignificant coefficient does not necessarily imply the absence of a causal relationship, because causal effects may be mediated, nonlinear, heterogeneous, or obscured by multicollinearity and limited variation.

3.4.5.1 Ordinary Least Squares (OLS)

To begin with, a cross-sectional OLS model is estimated on the 2025 snapshot of the data. Let **Data Centers** $_i$ denote the number of data centres in country i . The regression model is

$$\begin{aligned} \text{Data Centers}_i = & \beta_0 + \beta_1 \text{sustainability}_i + \beta_2 \text{incentives}_i + \beta_3 \text{planning}_i \\ & + \beta_4 \text{in_EU}_i + \beta_5 \text{reporting}_i + \beta_6 \text{Price (EUR/MWhe)}_i + \varepsilon_i. \end{aligned} \quad (3.36)$$

The OLS estimator is given by

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y, \quad (3.37)$$

where Y is the vector of data centre counts and X is the design matrix containing of regressors.

In this setting, the coefficient on each regressor measures the partial linear association between that variable and the number of data centres, conditional on the other regressors in the model. Because the regressors are selected using the causal DAG, the OLS specification is not interpreted as purely descriptive correlation, but as a causally informed linear approximation to the underlying structural relationships.

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Constant	-70.9922	107.340	-0.661	0.515	[-293.601, 151.616]
sustainability	4.9701	56.681	0.088	0.931	[-112.579, 122.519]
incentives	77.5263	29.245	2.651	0.015	[16.876, 138.176]
planning	46.1930	34.117	1.354	0.189	[-24.561, 116.947]
in_EU	-83.3646	52.582	-1.585	0.127	[-192.413, 25.683]
reporting	46.0942	54.856	0.840	0.410	[-67.669, 159.858]
Price (EUR/MWhe)	-0.0976	0.847	-0.115	0.909	[-1.855, 1.660]

Table 3.7: OLS regression results for the 2025 cross-section

The OLS regression explains a moderate proportion of the cross-country variation in data centre counts ($R^2 = 0.520$, adjusted $R^2 = 0.389$), and the model is jointly significant ($F = 3.976$, $p = 0.00762$). Among the included regressors, **incentives** is the only variable that is statistically significant at conventional levels, with a positive coefficient of 77.53 ($p = 0.015$). This suggests that incentive-based regulation is the clearest linear correlate of data centre deployment in the 2025 cross-section.

The remaining policy variables are not statistically significant: **sustainability** ($p = 0.931$), **planning** ($p = 0.189$), **in_EU** ($p = 0.127$), **reporting** ($p = 0.410$), and **Price (EUR/MWhe)** ($p = 0.909$). The negative coefficient on **in_EU** is therefore not strong evidence of a negative causal effect; rather, it indicates that once the other regressors are included, the direct linear association is weak and imprecisely estimated.

3.4.5.2 Causal-Aware Regression Models

In contrast with qualitative intuition, the causal DAG and the causal estimates indicate that the time-variant features, **Price (EUR/MWhe)**, **Renewables_Generation**, **Nuclear_Generation**, and **Fossil_Generation**, do not have significant causal impact on the growth of data centres. To supplement and verify this finding, a fixed-effects regression model was fit to estimate the correlation and significance of these time-variant features on the growth of data centres.

The panel nature of the dataset (country-year observations from 2023 to 2025) enables the use of fixed-effects regression to control for unobserved, time-invariant country characteristics such as baseline infrastructure maturity, historical policy environment, and cultural factors, known as **EntityEffects**. For treatments that vary over time, such as electricity price and generation mix, a causal-aware PanelOLS specification is employed.

Fixed-Effects Model. To account for the panel structure of the data, a fixed-effects model is estimated on the full country-year sample. Let **Data_Centers** $_{it}$ denote the number of data

centres in country i at time t . The fixed-effects model is

$$\begin{aligned} \text{Data_Centers}_{it} = & \alpha_i + \gamma_t + \theta_1 \text{Renewables_Generation}_{it} + \theta_2 \text{Nuclear_Generation}_{it} \\ & + \theta_3 \text{Fossil_Generation}_{it} + \theta_4 \text{Price (EUR/MWhe)}_{it} + u_{it}, \end{aligned} \quad (3.38)$$

where α_i is the country fixed effect, γ_t is the year fixed effect, and u_{it} is the idiosyncratic error term.

Variable	Coef.	Std. Err.	T-stat	p-value	95% CI
Renewables_Generation	-0.0003	0.0117	-0.0232	0.9816	[-0.0237, 0.0231]
Nuclear_Generation	0.0005	0.0035	0.1450	0.8853	[-0.0066, 0.0076]
Fossil_Generation	-0.0074	0.0077	-0.9624	0.3402	[-0.0228, 0.0080]
Price (EUR/MWhe)	0.0001	0.0058	0.0192	0.9847	[-0.0115, 0.0118]

Table 3.8: Fixed-effects panel regression results

The fixed-effects panel model has very little explanatory power ($R^2 = 0.0145$, within $R^2 = 0.0419$) and is not jointly significant ($p = 0.9398$). None of the time-varying energy variables are statistically distinguishable from zero: **Renewables_Generation** ($p = 0.9816$), **Nuclear_Generation** ($p = 0.8853$), **Fossil_Generation** ($p = 0.3402$), and **Price (EUR/MWhe)** ($p = 0.9847$). The coefficients are all numerically close to zero, indicating that there is little within-country variation over time in these variables that explains changes in data centre counts after accounting for country and year fixed effects.

The poolability test strongly rejects the pooled specification ($p < 0.0001$), which supports the use of fixed effects. At the same time, the near-zero within- R^2 shows that the short panel period provides limited identifying variation for these regressors.

Instrumental Variables. Instrumental variable (IV) regression was also considered, particularly for treatments such as electricity price that may be endogenous with respect to data centre deployment. However, the causal DAG and available data do not reveal any strong candidates for valid instruments that satisfy both relevance (strongly correlated with the treatment) and exclusion (affecting the outcome only through the treatment). Variables such as EU membership or generation mix directly influence data centre presence as well as price, violating the exclusion restriction. Therefore, IV estimation is not pursued as a primary identification strategy, and the analysis instead focuses on backdoor-adjusted OLS and fixed-effects regression supported by the DAG and DoWhy’s identification results.

3.4.6 Summary of Findings from Causal Analysis

Across the DoWhy causal estimands, several policy variables emerged as important causal drivers of data centre deployment. In particular, **planning** and **incentives** exhibited the strongest positive and statistically significant average treatment effects. **planning** had an estimated ATE of 82.71 with $p = 0.037$, while **incentives** had an estimated ATE of 56.97 with $p = 0.006$. **sustainability** also showed a positive and statistically significant estimated

effect ($ATE = 30.07$, $p = 0.040$). These results suggest that regulatory and policy design matters materially for the location and concentration of data centre activity, with incentive structures and planning frameworks appearing to be particularly influential.

By contrast, **Price** (EUR/MWhe) and **in EU** were associated with negative estimated effects, but these effects were not statistically significant in the causal estimands. This indicates that while electricity prices and institutional membership may shape the broader investment environment, their direct causal contribution is less clearly identified in the available sample once the adjustment set implied by the DAG is applied. Likewise, **reporting** showed a positive estimated effect, but with substantial uncertainty, suggesting that it may function more as a proxy for broader institutional maturity than as a strong standalone policy lever.

An especially notable result from the causal graph learning stage was that the generation variables (**Renewables_Generation**, **Nuclear_Generation**, and **Fossil_Generation**) did not retain a directed path to **Data_Centers** in the final adjusted DAG. As a consequence, DoWhy returned a zero causal effect for these variables under the learned graph. This does not necessarily mean that generation mix is irrelevant in reality; rather, it indicates that, given the current graph and the available data, no direct causal route from generation mix to data centre deployment was retained after structure learning and graph refinement.

3.4.6.1 Regression results and interpretation

The regression analyses complemented the causal estimators by providing linear summaries of the observed data. In the 2025 cross-sectional OLS model, **incentives** remained the clearest statistically significant regressor, while **planning** also showed a positive and significant coefficient in the more parsimonious specification. Population was highly significant and positive in all cross-sectional models, reinforcing the idea that larger countries and larger markets tend to host more data centres. However, several variables that were causally important in the DAG-based analysis were not strongly significant in the simple regression output.

This divergence is informative rather than contradictory. Causal significance does not require a large raw correlation in the observed data. A variable may exert a genuine causal effect while appearing weakly correlated if its effect is mediated by other variables, if the relationship is nonlinear, if the sample is small, or if the regressor is collinear with related policy and structural variables. This is particularly relevant here because the regulation variables are interdependent and the sample size is limited.

The fixed-effects panel regression further illustrated this distinction. Once country and year effects were included, the time-varying energy variables displayed almost no explanatory power for changes in data centre counts. This is consistent with the short time horizon of the panel and the fact that many of the most policy-relevant regulatory variables are time-invariant over 2023–2025 and are therefore absorbed by the fixed effects. In other words, the panel model is well suited to controlling for unobserved country heterogeneity, but it is not well suited to identifying the effects of static policy features in a short panel.

3.4.6.2 Causality versus correlation

One of the most important conceptual findings of the study is that causally significant variables do not always appear statistically correlated in a simple regression model. This distinction is central to causal inference. The causal graph encodes the direction of influence, while the regression coefficient reflects the strength of a linear conditional association in the sample. These are related but not equivalent.

For example, **sustainability** was significant in the DoWhy causal effect estimation, but was not always significant in the OLS specifications when entered jointly with other policy variables. Similarly, some variables with strong theoretical links to data centre growth, such as electricity generation mix or electricity price, had weak or null coefficients in the panel model. This may be because the effect operates indirectly, because the relevant variation is mostly between countries rather than within countries over time, or because the available data do not provide enough statistical power to isolate the effect cleanly.

This result should not be viewed as a failure of the model. Rather, it demonstrates why causal analysis requires more than regression alone. A variable can be causally relevant even if its simple correlation is weak, and a variable can be correlated with the outcome without representing a direct causal driver. The combination of DAG learning, backdoor adjustment, placebo refutation, OLS, and fixed-effects estimation therefore provides a more reliable and nuanced picture than any single method could offer on its own.

3.4.6.3 Geographical and temporal limitations

Finally, it is important to acknowledge several limitations that arise from the chosen approach for causal analysis, and the data supplemented. The two primary concerns that warrant an extended study, given more data and resources, are tied to the temporal limitations of the data, and the consequence that this has on analysing the impact of geographic and political factors within each country. Specifically, the 3 year time frame contained within the data does not encapsulate major changes in population size, or regulatory frameworks, which forced their exclusion from the fixed-effects model.

An extension to this study would warrant an analysis conducted over a wider time frame, which would encapsulate the growth of data centres over 5 to 10 years. This would be able to capture how that this growth impacted the changes and responses by regulatory frameworks, as new regulations typically take time to be put into effect. This would allow for a deeper understanding for the causal effect of these regulatory framework through the introduction of a temporal axis, and may give additional insight to this problem, beyond what has been covered in this study

4 Conclusion

Key Findings. The main practical implication of this analysis is that policy design appears to matter for data centre growth. In particular, planning frameworks and incentive schemes are the clearest causal levers identified. This was found in the network exposure regressions; incentives regulation emerging as a consistently significant factor with a positive effect on data centre presence ($p = 0.003 - 0.018$). Although the most extreme effect is driven solely by Ireland, our small sample size means this does not invalidate the finding, but rather warrants further research. All cross-sectional causal-aware regressions confirmed the positive significant effect of incentives regulation. Planning appears significant ($p = 0.037$) in it's DoWhy causal estimand, similarly exhibiting a positive effect on the presence of data centres. Causal analysis identifies sustainability regulation as relevant, though this is more indirect, intertwined with broader institutional context. These findings are directly relevant to policymakers seeking to attract data centre investment or manage its spatial and infrastructural consequences.

The ERGM shows a link between data centre presence and energy trade ties and a highly significant relationship between trade and price differences between countries. The Network Autocorrelation model confirms that countries' prices are not independent of one another ($LR = 12.9 - 21.6$, $p < 0.001$), whilst US data linear regressions showed that price is a significant predictor of data centres ($p = 0.011 - 0.012$). Although the coefficient sign flips upon controlling for population via a per-capita data centre measure, the significance of this relationship is the key insight here. Holistically, these findings indicate that countries linked by energy trade affect each other's prices, which we found separately to have a relationship with data centres; indirectly linking our key variables. The causal mechanisms involved in this warrant further empirical study, presenting opportunity for further research.

Contributions. We have shown how a causally informed pipeline can be built from observational panel data even when treatment variables are time-invariant and the outcome is shaped by a mixture of factors. This may be extended to other infrastructure- and policy-sensitive sectors where the goal is not simply to predict an outcome, but to understand which levers are capable of changing it. Furthermore, to our knowledge, ERGMs have not been applied to energy trade networks and data centres. This shows how horizontal application of data science techniques can reveal new insights; we welcome review and criticism of this novel approach.

Extensions. On the modelling side, future work could employ double machine learning or causal forests to capture non-linear and heterogeneous treatment effects while retaining DAG-informed adjustment sets. On the data side, extending the panel to a longer time horizon and incorporating more granular measures of variables would enable richer causal identification, possibly including the use of instrumental variables based on exogenous shocks or historical policy discontinuities. Crucially, further study considering a wider global scope would aid in better holistic understanding of this highly complex and fast-growing problem.

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